

FINAL PROJECT - ES234849

HIERARCHICAL CLUSTER ANALYSIS OF MEDICAL STUDENT PERFORMANCE ACROSS MULTIPLE ASSESSMENT MODALITIES

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Institut Teknologi Sepuluh Nopember

Surabaya

2026



TUGAS AKHIR - ES234849

ANALISIS KLASSTER TERHADAP KINERJA MAHASISWA KEDOKTERAN PADA BERBAGAI MODALITAS PENELITIAN

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APPROVAL SHEET

HIERARCHICAL CLUSTER ANALYSIS OF MEDICAL STUDENT PERFORMANCE ACROSS MULTIPLE ASSESSMENT MODALITIES

FINAL PROJECT

Submitted to fulfill one of the requirements
for obtaining a degree S.Kom at
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Faculty of Electrical Engineering and Intelligent Computing
Institut Teknologi Sepuluh Nopember

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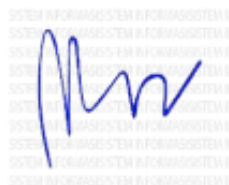
hereby declare that the Final Project with the title
"HIERARCHICAL CLUSTER ANALYSIS OF MEDICAL
STUDENT PERFORMANCE ACROSS MULTIPLE
ASSESSMENT MODALITIES" is the result of my own work,
is original, and is written by following the rules of scientific
writing.

If in the future there is a discrepancy with this statement, then I am willing
to accept sanctions in accordance with the provisions that apply at Institut
Teknologi Sepuluh Nopember.

Surabaya, 16 Juli 2026

Acknowledged

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ABSTRAK

HIERARCHICAL CLUSTER ANALYSIS OF MEDICAL STUDENT PERFORMANCE ACROSS MULTIPLE ASSESSMENT MODALITIES

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Abstrak

Penilaian pendidikan kedokteran bersifat kompleks karena melibatkan berbagai modalitas ujian seperti teoritertulis, keterampilan praktik, pasien terstandarisasi, dan pembelajaran berbasis masalah. Metode statistik konvensionalsering mengevaluasi setiap modalitas secara terpisah, sehingga mengaburkan pola kinerja lintas-modalitas. Penelitian inimenerapkan analisis klaster hierarkis untuk mengidentifikasi profil kinerja mahasiswa kedokteran di Fakultas Kedokteran dan Kesehatan (FKK) ITS Surabaya. Data dikumpulkan dari 98 mahasiswa angkatan 2023 dan 2024 yang memiliki nilai lengkap pada empat modalitas. Metode Ward dengan jarak Euclidean digunakan setelah standardisasi Z-score. Hasilpenelitian menghasilkan tiga klaster yang berbeda secara statistik (ANOVA, $p < 0,001$): Klaster 1 (At-Risk, 26,5%) dengandefisit luas di tiga modalitas; Klaster 2 (Mixed Performance, 9,2%) dengan keunggulan di ujian terstruktur tetapi keagalankatastropik pada problem-based ($Z = -2,236$); Klaster 3 (Average, 64,3%) dengan kinerja seimbang di atas rata-rata. Koefisien Silhouette 0,3913 dan Cophenetic Correlation 0,6401 menunjukkan pemisahan klaster moderat namun dapat diterima. Temuan ini mendukung intervensi pendidikan yang ditargetkan, optimalisasi strategi penilaian, dan kerangka identifikasidini mahasiswa berisiko. Metodologi ini bersifat reproduibel dan dapat diadopsi institusi pendidikan kedokteran lain.

Kata kunci: *hierarchical clustering, medical education, multimodal assessment, student performance, Ward's method, educational data mining*

ABSTRACT

HIERARCHICAL CLUSTER ANALYSIS OF MEDICAL STUDENT PERFORMANCE ACROSS MULTIPLE ASSESSMENT MODALITIES

Student Name / NRP : Gerald Mahapranaja Pillian / 5026221050
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Abstract

Medical education assessment is complex because it involves various examination modalities such as written theory, practical skills, standardised patients, and problem-based learning. Conventional statistical methods often evaluate each modality separately, thus obscuring cross-modality performance patterns. This study applies hierarchical cluster analysis to identify the performance profiles of medical students at the Faculty of Medicine and Health (FKK) ITS Surabaya. Data were collected from 98 students from the 2023 and 2024 cohorts who had complete scores across four modalities. Ward's method with Euclidean distance was applied after Z-score standardisation. The study yielded three statistically distinct clusters (ANOVA, $p < 0.001$): Cluster 1 (At-Risk, 26.5%) with broad deficits across three modalities; Cluster 2 (Mixed Performance, 9.2%) with strengths in structured examinations but catastrophic failure in problem-based assessment ($Z = 2.236$); and Cluster 3 (Average, 64.3%) with balanced above-average performance. A Silhouette Coefficient of 0.3913 and Cophenetic Correlation of 0.6401 indicate moderate but acceptable cluster separation. These findings support targeted educational interventions, optimisation of assessment strategies, and an early identification framework for at-risk students. The methodology is reproducible and can be adopted by other medical education institutions.

Keywords: *hierarchical clustering, medical education, multimodal assessment, student performance, Ward's method, educational data mining*

PREFACE

First and foremost, I would like to express my gratitude to God for the strength, guidance, and grace that allowed me to complete this undergraduate thesis, titled "Hierarchical Cluster Analysis of Medical Student Performance Across Multiple Assessment Modalities," submitted in fulfilment of the requirements for the degree of Bachelor of Science (S-1) in the Department of Information Systems, Faculty of Electrical Technology and Intelligent Informatics (FTEIC), Institut Teknologi Sepuluh Nopember (ITS), Surabaya.

Throughout this process, I received a great deal of support and assistance, both direct and indirect, from many people. I would like to extend my sincere thanks to:

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I am aware that this thesis is still far from perfect. I therefore welcome any feedback and suggestions that could help improve this research further. I hope that this thesis will be of benefit both to myself and to its readers.

Surabaya, 27 July 2026

The Author

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CHAPTER I INTRODUCTION

In this chapter, the author will introduce the subject of the undergraduate thesis, which is the hierarchical cluster analysis of medical student performance across multiple assessment modalities. This chapter contains the discussion of the background, problem formulation, scope, objectives, and benefits of this undergraduate thesis. The background section provides an overview of the medical education assessment problem, the existing solutions to the problem of students' performance evaluation, and a brief explanation of clustering techniques typically applied to educational data analysis.

1.1 Background

Medical education assessment is complex, placing a significant responsibility on educators and medical administrators to understand student performance patterns (Eva, 2009). Traditional methods of performance evaluation often assess each modality independently, which means the connections between different forms of learning and performance may be overlooked (Watling & Ginsburg, 2019). Different students can demonstrate varying levels of competency across assessment types, with some performing well in theoretical examinations while struggling in practical assessments (Norman et al., 2014).

Assessment complexity intensifies due to medical education's simultaneous demands: formative evaluation for monitoring developmental progress, summative evaluation for advancement determinations, and diagnostic evaluation for pinpointing learning deficiencies (Schuwirth & Van der Vleuten, 2011). The volume of assessment data generated across diverse modalities makes it difficult for educators to identify meaningful patterns in student performance without the use of advanced analytical approaches (Rienties et al., 2016). This challenge is further complicated by the variation in medical student populations in terms of educational background, learning styles, and career goals (Artino et al., 2014).

To address these analytical challenges, clustering analysis has emerged as one of the most effective approaches for uncovering complex patterns in educational data (Romero & Ventura, 2020). Clustering algorithms can identify groups of students that exhibit similar performance patterns across multiple assessment methods (Baker & Yacef, 2009). Hierarchical clustering is particularly valuable in educational contexts because it provides a visual and interpretable structure of student groupings at various levels of similarity, enabling educators to capture both broad performance clusters and finer performance differences (Aggarwal & Reddy, 2014).

Building on this methodological foundation, the application of hierarchical clustering to medical education assessment can reveal latent performance patterns that would not be apparent through conventional analysis methods (Jain et al., 1999). This approach is useful in identifying students who may benefit from targeted interventions, uncovering assessment modalities that measure similar underlying competencies, and informing curriculum design decisions (Han et al., 2012). Research has demonstrated that clustering can help identify at-risk students earlier than traditional approaches, enabling timely and effective intervention (Peña-Ayala, 2014).

Based on these considerations, this undergraduate thesis proposes applying hierarchical cluster analysis to gain insight into student performance patterns across multiple assessment modalities, with the goal of supporting more tailored educational interventions and enhanced evaluation practices. The medical education context in Surabaya, Indonesia provides a compelling setting for this research. Medical institutions in Surabaya, including those affiliated with Institut Teknologi Sepuluh Nopember (ITS), combine structured academic curricula with

clinical training, generating rich multimodal assessment data that is well-suited to hierarchical clustering analysis.

1.2 Problem Formulation

Based on the background that has been described, the following research questions are formulated for this thesis:

1. How can a hierarchical cluster analysis be effectively applied to analyse medical student performance across multiple assessment modalities?
2. How do the identified clusters relate to student characteristics and learning outcomes in medical education?
3. What distinct performance patterns emerge when medical students are clustered based on their scores across different assessment types?
4. What insights can hierarchical clustering provide for improving assessment strategies and educational interventions in medical education?

1.3 Scope

The scope of this undergraduate thesis is limited to the following parameters:

1. **Participants:** Medical students enrolled in clinical or pre-clinical programs at medical institutions in Surabaya, Indonesia during the academic year 2025/2026.
2. **Assessment Modalities:** The study will focus on commonly used assessment methods in medical education including:
 - a. Written theoretical examination
 - b. Practical skill assessments
 - c. Standardised patient encounters
 - d. Problem-based assessments
3. **Analytical Method:** Hierarchical cluster analysis will be the primary analytical technique, with consideration of different linkage methods and distance measures appropriate for educational data.

1.4 Objectives

Based on the problem formulation explained above, the objectives of this thesis are:

1. **Primary objective:** To use hierarchical cluster analysis to identify particular patterns of performance among medical students in various types of evaluation.
2. **Secondary objectives:**
 - a. Describe the discovered student clusters in terms of their performance profiles in the various types of evaluation.
 - b. To analyse the relationship between cluster membership and demographic or academic attributes of the students.
 - c. To evaluate the practical impact of clustering outcomes on medical training.

- d. To provide advice on how to optimise assessment practice in accordance with the performance patterns discovered.

1.5 Contributions

This research contributes both theoretical insights and practical applications:

1.5.1 Theoretical Contribution

1. **Methodological Development:** Expands the application of hierarchical clustering methods in medical education assessment and contributes to the knowledge base of educational data mining in healthcare education contexts.
2. **Understanding Performance Trends:** Provides information on how medical students perform in various assessment modalities and identifies common performance trends among the medical student community.
3. **Assessment Relationship Analysis:** Using cluster analysis to investigate the relationships between different assessment modalities, which contributes to our understanding of assessment validity and reliability in medical education.

1.5.2 Practical Contribution

1. **Educational Intervention Design:** The clustering results can be used to create tailored education interventions for different student performance profiles, potentially improving learning outcomes and student support services.
2. **Assessment Strategy Optimisation:** Providing evidence-based guidance on maximising assessment strategies in medical education, such as redundancy or complementary testing modalities.
3. **Early Identification System:** Creating a system/framework for identifying students who can benefit from extra support or other learning methods based on their performance cluster features.
4. **Institutional Decision Support:** Providing evidence-based information to inform curriculum creation, resource management, and student advisory decisions at medical education institutions.

Replicable Framework: Creating a reproducible analytic framework that can be updated and used in other medical education institutions or similar healthcare teaching circumstances.

CHAPTER II LITERATURE REVIEW

In this chapter, the authors provide a comprehensive literature review and theoretical foundation for hierarchical cluster analysis in medical education assessment. The literature study examines previous research on medical student performance assessment, hierarchical clustering approaches in education, and multimodal assessment of medical education. The theoretical backdrop provides the conceptual context for applying hierarchical clustering methods to data on medical student performance.

2.1 Previous Studies

In recent years, clustering methods have proven useful in establishing patterns in student performance across several educational domains.

Zhang et al. (2021) systematically examined educational data mining methodologies for predicting student outcomes, encompassing clustering, classification, and regression techniques. Their analysis validated clustering's capacity to aggregate students with comparable academic trajectories, with hierarchical approaches yielding particularly interpretable groupings suited to personalized instructional planning. The study highlighted that conventional grading approaches frequently underestimate the complexity of student learning patterns, and that more advanced analytical methods are needed.

Qadir, Meghji, and Mahoto (2025) employed data-driven clustering to track academic progression throughout a complete degree program. Analysing 210 student records via k-means and hierarchical methods, they distinguished three performance tiers—high, intermediate, and low achievers—with heatmap analysis revealing subject-specific performance variation. Their findings demonstrated that clustering reveals performance patterns that simple average-based groupings miss entirely.

Lu, Y., Yeom, S., Maktoubian, J., et al. (2025) conducted a systematic review of cluster-based student risk prediction models, examining 61 studies published between 2010 and 2025. They found that hierarchical clustering has been widely applied to organise students based on academic performance and learning behaviours, and that it helps reduce noise and improves the precision of at-risk identification models. The review confirmed that clustering-based models capture behavioural and academic differences that enable institutions to provide more adaptive and timely support.

Okubo et al. (2023) provided a thorough review of clustering models applied in educational data science, analysing over 133 publications from 1983 onwards. They found a consistent upward trend in the use of clustering in education research, with hierarchical clustering being among the most frequently applied methods. The review also noted that appropriate data preprocessing and feature selection are critical to obtaining meaningful clustering results.

Ward's linkage criterion has been established as one of the most effective approaches for hierarchical clustering of multidimensional data, as it minimises within-cluster variance at each merging step and produces well-balanced, interpretable clusters (Ward, 1963). This property is directly relevant to the medical education context, where assessment data combines quantitative scores with qualitative evaluations across multiple modalities, and where cluster interpretability is essential for informing educational decisions (Jain et al., 1999).

Peña-Ayala (2014) conducted a landmark survey of educational data mining, finding that clustering is effective across a wide range of educational environments and that it can help identify at-risk students earlier than traditional approaches, allowing interventions to be implemented timelier. The study also highlighted that clustering analysis can guide individualised learning and inform curriculum design decisions.

Romero and Ventura (2020) provided an updated and comprehensive survey of educational data mining and learning analytics, reinforcing the finding that clustering methods consistently outperform traditional statistical methods in uncovering hidden patterns in student performance data, particularly in diverse and multicultural educational settings.

Overall, the extensive body of prior research confirms that hierarchical cluster analysis is a promising and well-validated method for analysing medical student performance across multiple assessment modalities. The literature consistently demonstrates the usefulness of hierarchical clustering in extracting meaningful patterns from student performance data, particularly in diverse educational contexts such as medical programmes in Indonesia.

2.2 Theoretical Background

The theoretical foundation for this study encompasses several key areas: medical education assessment theory, cluster analysis methodology, and multi-modal learning evaluation frameworks. This section provides the conceptual basis for applying hierarchical clustering techniques to medical student performance analysis.

2.2.1 Medical Education Assessment

Medical education evaluation is a well-developed assessment method designed to evaluate students' skills and performance in areas such as knowledge, competency, and professional attitude (Miller, 1990). The assessment system in medical education is mostly based on Miller's Pyramid, which starts with the proverbial knowledge pyramid, beginning with the most basic level of knowledge, that is “knows” (factual knowledge). It then progresses to “knows how” (applied knowledge), followed by “shows how” (demonstrated competence), and ends with the highest level, “does”, which reflects performance in real clinical practice (Norcini & McKinley, 2007).

ALGORITHM 1: Medical Student Performance Pattern Analysis

INPUT: Cluster profiles P , Student data S , Assessment weights W

OUTPUT: Performance patterns T , Risk assessment R

```

1:  PROCEDURE PerformancePatternAnalysis( $P$ ,  $S$ ,  $W$ )
2:     $k \leftarrow \text{number\_of\_clusters}(P)$ 
3:     $T \leftarrow \text{empty\_dictionary}()$ 
4:     $R \leftarrow \text{empty\_array}(\text{number\_of\_students}(S))$ 
5:
6:    // Identify performance patterns for each cluster
7:    FOR cluster_id = 1 to  $k$  DO
8:      cluster_profile  $\leftarrow P[\text{cluster\_id}, :]$ 
9:
10:     // Calculate weighted performance score
11:     weighted_score  $\leftarrow \text{sum}(\text{cluster\_profile} \times W) / \text{sum}(W)$ 
12:
13:     // Identify strengths and weaknesses
14:     strengths  $\leftarrow \text{find\_indices\_above\_threshold}(\text{cluster\_profile}, 0.5)$ 

```

```

15:     weaknesses ← find_indices_below_threshold(cluster_profile, -0.5)
16:
17:     // Categorize performance pattern
18:     IF weighted_score > 0.5 THEN
19:         pattern_type ← "High Achiever"
20:     ELSE IF weighted_score < -0.5 THEN
21:         pattern_type ← "At-Risk"
22:     ELSE IF variance(cluster_profile) > 1.0 THEN
23:         pattern_type ← "Mixed Performance"
24:     ELSE
25:         pattern_type ← "Average"
26:     END IF
27:
28:     T[cluster_id] ← {
29:         "type": pattern_type,
30:         "weighted_score": weighted_score,
31:         "strengths": strengths,
32:         "weaknesses": weaknesses,
33:         "profile": cluster_profile
34:     }
35:     END FOR
36:
37:     // Individual student risk assessment
38:     FOR student_id = 1 to number_of_students(S) DO
39:         student_cluster ← S[student_id]["cluster"]
40:         cluster_pattern ← T[student_cluster]
41:
42:         // Risk score calculation
43:         IF cluster_pattern["type"] == "At-Risk" THEN
44:             R[student_id] ← 0.8 + 0.2 × random()
45:         ELSE IF cluster_pattern["type"] == "Mixed Performance" THEN
46:             weakness_severity ← mean(cluster_pattern["weaknesses"])
47:             R[student_id] ← 0.3 + 0.4 × weakness_severity
48:         ELSE
49:             R[student_id] ← 0.1 + 0.2 × random()
50:         END IF
51:     END FOR
52:
53:     RETURN T, R
54: END PROCEDURE

```

(Adapted from Peña-Ayala, 2014; Romero & Ventura, 2020)

Program Code 1. Student Performance Pattern Analysis

Modern medical education emphasises competency-based assessment, which is based on individual clinical results rather than time (Frank et al., 2010). This technique demands the use of modern analytical tools to simultaneously examine the results of many competency areas for pupils. Common methods of evaluation in medical education include written tests, practical skill assessments, clinical assessments, and portfolio-based assessments (Schuwirth & Van der Vleuten, 2011).

The assessment of medical education is complex, making it difficult to understand statistics about student performance. Conventional statistical methods are ineffective in

identifying the multifaceted nature of medical skill (Eva, 2018). As a result of this constraint, there has been increased interest in different analytical methodologies such as clustering analysis to gain a better understanding of student learning habits and performance profiles.

2.2.2 Cluster Analysis in Educational Research

Cluster analysis represents a machine learning methodology whereby naturally occurring groups emerge from similarity patterns within datasets (Hastie, Tibshirani, & Friedman, 2021). Clustering techniques have become more important in educational research as a means of identifying similarities in students' behaviours, performance, and learning patterns.

Cluster analysis, when used in schools, can offer a number of advantages, including the ability to identify subgroups of students based on similar features, create individualised learning approaches, or provide educational interventions. According to Peña-Ayala (2014), educational clustering is typically used to improve academic achievement, learning behaviours, and engagement patterns.

2.2.3 Hierarchical Clustering Methodology

Hierarchical clustering constructs tree-like grouping structures through iterative merging or partitioning operations based on proximity measures. Hierarchical clustering is classified into two types: agglomerative (bottom-up) and divisive (top-down) (Rokach and Maimon, 2005).

Agglomerative approaches initiate with each data point as an individual cluster, progressively combining proximate groups until achieving a unified structure (Everitt et al., 2011). The procedure generates a dendrogram, which visually illustrates the hierarchical distribution of the data, allowing the researcher to determine the optimal number of clusters to react with in order to meet the analysis's aims and objectives.

ALGORITHM 4: Ward Hierarchical Clustering

INPUT: Standardized data matrix $X[n \times m]$, linkage_method

OUTPUT: Cluster hierarchy H , distance_matrix D

```

1:  PROCEDURE WardHierarchicalClustering( $X$ , linkage_method)
2:    // Initialize distance matrix
3:     $D \leftarrow \text{zeros}(n \times n)$ 
4:    FOR  $i = 1$  to  $n$  DO
5:      FOR  $j = i+1$  to  $n$  DO
6:         $D[i,j] \leftarrow \text{euclidean\_distance}(X[i,:], X[j,:])$ 
7:         $D[j,i] \leftarrow D[i,j]$ 
8:      END FOR
9:    END FOR
10:
11:    // Initialize clusters (each point as separate cluster)
12:    clusters  $\leftarrow \{C_1, C_2, \dots, C_n\}$  where  $C_i = \{i\}$ 
13:     $H \leftarrow \text{empty\_hierarchy}()$ 
14:
15:    WHILE  $|\text{clusters}| > 1$  DO
16:      // Find closest cluster pair
17:      min_distance  $\leftarrow \infty$ 
18:      merge_i, merge_j  $\leftarrow \text{null}, \text{null}$ 
19:
20:      FOR each pair  $(C_i, C_j)$  in clusters WHERE  $i \neq j$  DO
21:        IF linkage_method = "ward" THEN

```

```

22:     distance ← ward_distance(Ci, Cj, X)
23:     ELSE IF linkage_method = "complete" THEN
24:         distance ← complete_linkage(Ci, Cj, D)
25:     ELSE IF linkage_method = "average" THEN
26:         distance ← average_linkage(Ci, Cj, D)
27:     END IF
28:
29:     IF distance < min_distance THEN
30:         min_distance ← distance
31:         merge_i, merge_j ← i, j
32:     END IF
33: END FOR
34:
35: // Merge closest clusters
36: C_new ← Cmere_i ∪ Cmere_j
37: clusters ← clusters - {Cmere_i, Cmere_j} + {C_new}
38: H.add_merge(merge_i, merge_j, min_distance)
39: END WHILE
40:
41: RETURN H, D
42: END PROCEDURE

```

(Adapted from Garcia et al., 2016; Kotsiantis et al., 2007)

Program Code 2. Ward Hierarchical Clustering for Educational Data ALGORITHM

The distance measure and the linking requirement have a significant impact on the clustering results (Müllner, 2013). Distance measures include Euclidean distance, Manhattan distance, and correlation-based distances, whereas linkage criteria include single linkage, complete linkage, average linkage, and Ward (James et al., 2021).

ALGORITHM 3: Ward Distance Calculation

INPUT: Cluster C_i, Cluster C_j, Data matrix X

OUTPUT: Ward distance between clusters

```

1:  FUNCTION ward_distance(Ci, Cj, X)
2:    // Calculate cluster centroids
3:    μi ← mean(X[Ci, :]) // Centroid of cluster i
4:    μj ← mean(X[Cj, :]) // Centroid of cluster j
5:
6:    // Calculate merged cluster centroid
7:    ni ← |Ci| // Size of cluster i
8:    nj ← |Cj| // Size of cluster j
9:    μmere ← (ni × μi + nj × μj) / (ni + nj)
10:
11:   // Calculate within-cluster sum of squares before merge
12:   SSW_before ← 0
13:   FOR each point p in Ci DO
14:     SSW_before ← SSW_before + ||X[p,:] - μi||2
15:   END FOR
16:   FOR each point p in Cj DO
17:     SSW_before ← SSW_before + ||X[p,:] - μj||2
18:   END FOR
19:
20:   // Calculate within-cluster sum of squares after merge
21:   SSW_after ← 0
22:   FOR each point p in (Ci ∪ Cj) DO

```

```

23:     SSW_after ← SSW_after + ||X[p,:] - μmere||2
24: END FOR
25:
26: // Ward distance is the increase in SSW
27: ward_dist ← SSW_after - SSW_before
28:
29: RETURN ward_dist
30: END FUNCTION

```

(Adapted from Ward, 1963; Jain et al., 1999)

Program Code 3. Ward Distance Calculation

Ward's approach, which involves minimising within-cluster variations, has been demonstrated to be far superior when dealing with educational data clustering applications (Steinley 2006). Compared to all other linkage methods, this method is more prone to producing dense spherical clusters and is more resistant to outliers.

2.2.4 Multi-Modal Assessment in Medical Education

Multi-modal assessment is a term used to define multi-modal review of student competency (Wass et al., 2011). This methodology recognizes that diverse assessment formats capture distinct dimensions of student learning and competency (Driessen et al., 2007).

When several modalities are included, analytical problems arise, since the various types of assessments will often differ in scale, distribution as well as measurement properties (Govaerts & van der Vleuten, 2013). Clustering analysis provides an encouraging method of combining the data obtained in different forms of assessments to develop a very broad general picture of student performance.

2.2.5 Data Preprocessing for Educational Clustering

García, Luengo, & Herrera (2016) found that clustering educational data can yield valuable results with proper preparation. Some of the most essential preprocessing operations include missing data processing, standardisation of different scaled variables, and feature selection (Kotsiantis et al., 2007).

ALGORITHM 2: Medical Assessment Data Preprocessing

INPUT: Raw assessment data matrix $X[n \times m]$, missing_threshold, outlier_threshold

OUTPUT: Preprocessed data matrix $X_processed[n \times m]$

```

1: PROCEDURE DataPreprocessing(X, missing_threshold, outlier_threshold)
2: // Handle Missing Values
3: FOR each column j in X DO
4:     missing_ratio ← count_missing(X[:,j]) / n
5:     IF missing_ratio > missing_threshold THEN
6:         remove_column(X, j)
7:     ELSE
8:         X[:,j] ← impute_mean(X[:,j])
9:     END IF
10: END FOR
11:
12: // Detect and Handle Outliers
13: FOR each row i in X DO
14:     z_scores ← calculate_z_scores(X[i,:])

```

```

15:     IF max(abs(z_scores)) > outlier_threshold THEN
16:         X[i,:] ← winsorize(X[i,:], limits= [0.05, 0.95])
17:     END IF
18: END FOR
19:
20: // Standardised Features
21: FOR each column j in X DO
22:      $\mu_j \leftarrow \text{mean}(X[:,j])$ 
23:      $\sigma_j \leftarrow \text{standard\_deviation}(X[:,j])$ 
24:     X[:,j] ← (X[:,j] -  $\mu_j$ ) /  $\sigma_j$ 
25: END FOR
26:
27: RETURN X
28: END PROCEDURE

```

(Adapted from Ward, 1963; Jain et al., 1999)

Program Code 4. Data Preprocessing for Medical Assessment Data

Preprocessing elements in medical education contexts include the incorporation of varied evaluation weightings, longitudinal data structures, and data from distinct student cohorts Baker, R. S. (2010). Adequate preprocessing is required to ensure that clustering outcomes may be related to and used by educational stakeholders.

2.2.6 Evaluation Methods for Educational Clustering

Clustering findings in educational contexts should be assessed or appraised using both statistical validity and educational significance (Jain, 2010). Statistical assessment tools include silhouette, within-cluster sum of squares, and cophenetic correlation coefficient (Rousseeuw, 1987).

ALGORITHM 5: Optimal Cluster Number Using Multiple Criteria

INPUT: Data matrix X, Hierarchy H, max_clusters

OUTPUT: Optimal number of clusters k_opt

```

1:  PROCEDURE OptimalClusterNumber(X, H, max_clusters)
2:      silhouette_scores ← empty_array(max_clusters-1)
3:      calinski_scores ← empty_array(max_clusters-1)
4:      davies_bouldin_scores ← empty_array(max_clusters-1)
5:      elbow_scores ← empty_array(max_clusters-1)
6:
7:      FOR k = 2 to max_clusters DO
8:          // Extract k clusters from hierarchy
9:          clusters ← cut_dendrogram(H, k)
10:         labels ← assign_cluster_labels(X, clusters)
11:
12:         // Calculate silhouette score
13:         silhouette_scores[k-2] ← calculate_silhouette(X, labels)
14:
15:         // Calculate Calinski-Harabasz index
16:         calinski_scores[k-2] ← calculate_calinski_harabasz(X, labels)
17:
18:         // Calculate Davies-Bouldin index
19:         davies_bouldin_scores[k-2] ← calculate_davies_bouldin(X, labels)

```

```

20:
21:     // Calculate within-cluster sum of squares (for elbow method)
22:     elbow_scores[k-2] ← calculate_wcss(X, labels)
23: END FOR
24:
25:     // Find optimal k using multiple criteria
26:     k_silhouette ← argmax(silhouette_scores) + 2
27:     k_calinski ← argmax(calinski_scores) + 2
28:     k_davies ← argmin(davies_bouldin_scores) + 2
29:     k_elbow ← find_elbow_point(elbow_scores) + 2
30:
31:     // Consensus decision (mode of the results)
32:     k_candidates ← [k_silhouette, k_calinski, k_davies, k_elbow]
33:     k_opt ← mode(k_candidates)
34:
35:     RETURN k_opt
36: END PROCEDURE

```

(Adapted from Tibshirani et al., 2001; Rousseeuw, 1987)

Program Code 5. Optimal Cluster Number Determination

Meaningful education evaluation necessitates an examination of the significance of the specialist groups revealed by devices in terms of educationally relevant student grouping, as well as how the clustering output may be used in education (Perera et al. 2009). This simple two-step evaluation approach guarantees the statistical and educational significance of clustering results.

ALGORITHM 6: Cross-Validation for Clustering Stability

INPUT: Data matrix X, Number of folds cv_folds, Number of iterations n_iter

OUTPUT: Stability score stability_mean, Confidence interval CI

```

1: PROCEDURE CrossValidationStability(X, cv_folds, n_iter)
2:     stability_scores ← empty_array(n_iter)
3:     n ← number_of_rows(X)
4:
5:     FOR iteration = 1 to n_iter DO
6:         fold_scores ← empty_array(cv_folds)
7:
8:         // Create stratified folds
9:         folds ← create_stratified_folds(X, cv_folds)
10:
11:        FOR fold = 1 to cv_folds DO
12:            // Split data
13:            X_train ← X[folds != fold, :]
14:            X_test ← X[folds == fold, :]
15:
16:            // Perform clustering on training set
17:            H_train ← hierarchical_clustering(X_train, method="ward")
18:            k_opt ← determine_optimal_clusters(X_train, H_train)
19:            clusters_train ← cut_dendrogram(H_train, k_opt)
20:

```

```

21:      // Predict cluster assignments for test set
22:      clusters_test ← predict_cluster_assignments(X_test, clusters_train)
23:
24:      // Calculate cluster stability (Adjusted Rand Index)
25:      true_clusters_test ← get_true_clusters(X_test, H_train, k_opt)
26:      fold_scores[fold] ← adjusted_rand_index(clusters_test, true_clusters_test)
27:  END FOR
28:
29:      stability_scores[iteration] ← mean(fold_scores)
30:  END FOR
31:
32:  // Calculate statistics
33:  stability_mean ← mean(stability_scores)
34:  stability_std ← standard_deviation(stability_scores)
35:  CI ← [stability_mean - 1.96×stability_std, stability_mean + 1.96×stability_std]
36:
37:  RETURN stability_mean, CI
38:  END PROCEDURE

```

(Adapted from Halkidi et al., 2001; Kaufman & Rousseeuw, 2009)

Program Code 6. Cross-Validation for Clustering Stability

The current theoretical foundations proposed in this literature review identify the source from which a concept such as hierarchical clustering analysis could be applied to medical students' performance data in a variety of evaluation modalities. Combining medical education evaluation theory with a sophisticated clustering algorithm yields a potentially useful method for making sense of complex structures of student learning and performance.

2.2.7 Evaluation Methods for Educational Clustering

Medical education assessment in Indonesian programs employs specialized measurement frameworks that differ fundamentally from conventional academic evaluation systems. Unlike traditional GPA-based assessments, medical education integrates competency-based frameworks (Frank et al., 2010), clinical skill evaluations, and professional behavior metrics that require domain-specific interpretation.

Indonesian medical programs, particularly those following national standards set by the Ministry of Education and professional medical bodies, utilize multimodal assessment approaches including:

1. Competency-based progression criteria
2. Objective Structured Clinical Examinations (OSCE)
3. Standardized patient encounters with specialized rubrics
4. Professional behaviour and attitude assessments

These assessment modalities may not follow normal statistical distributions and often employ threshold-based pass/fail criteria rather than continuous grading scales (Norcini & McKinley, 2007). When applying clustering analysis to medical education data, it is essential to validate that statistical patterns align with educationally meaningful performance categories and respect the unique structure of competency-based assessment frameworks.

CHAPTER III METHODOLOGY

This chapter describes the research methodology used in this study to analyse medical students' performance in various types of assessments using hierarchical cluster analysis. The approach will comprise study design, data gathering techniques, an analytical framework, and assessment methodologies for achieving the research objectives. Figure 3.1 illustrates the study method's thorough flowchart, which explains the investigation.

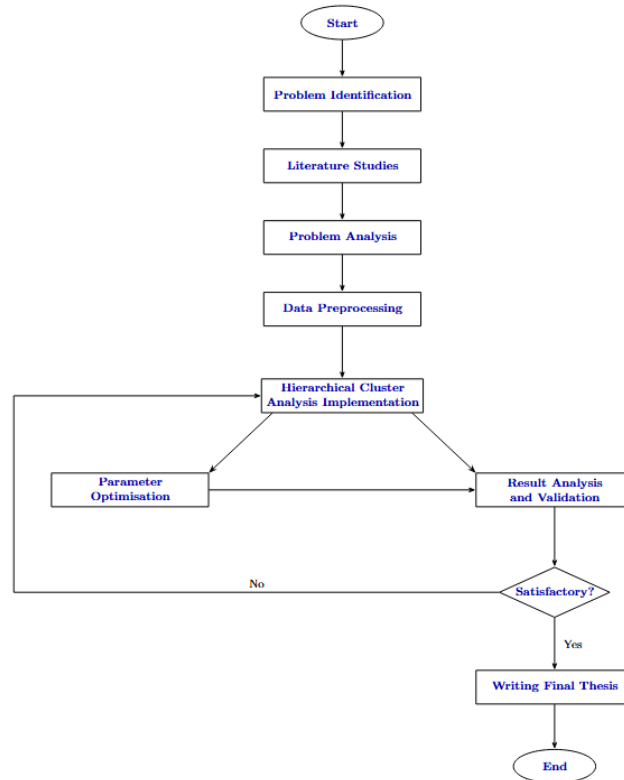


Figure 3.1 Research Methodology Flowchart

3.1 Problem Identification

The first stage of research is to identify the problem, which includes establishing research questions, determining the study scope, defining research goals, and formulating expected contributions. Educational data mining has become a significant topic due to the need to understand complex learning patterns through thorough analytical approaches (Romero & Ventura, 2020). The identification technique entails creating appropriate clustering procedures based on medical education data and developing evaluation models that can be used to assess student performance.

In this study, the researchers aimed to compare the performance of medical student groups at the Fakultas Kedokteran dan Kesehatan (FKK), Institut Teknologi Sepuluh Nopember (ITS), Surabaya, across various modes of assessment, such as theoretical examinations, practical tests, clinical tests, and research projects. The problem analysis is concerned with the need to implement more sophisticated analytical tools that can reveal implicit patterns in student performance data that conventional statistical methods cannot effectively access (Baker & Inventado, 2014).

The target population of this study is first- to fourth-year students pursuing medicine during the 2025/2026 academic year. The specific focus of the study is to define performance clusters that could be used to understand individual learning needs and assist in designing learning interventions and curricular approaches.

3.2 Literature Studies

This literature synthesis establishes theoretical grounding by examining existing scholarship on educational clustering applications, medical education evaluation frameworks, and student performance analysis methodologies Peña-Ayala (2014) analysed peer-reviewed journal publications, conference reports, and dissertations on educational data mining and clustering in medical education.

The literature study will focus on identifying the topics of research discovered in the present clustering methodologies utilised in medical education, specifically regarding multi-modal assessment data integration. The utility of hierarchical clustering in distinguishing student learning patterns has previously been demonstrated in various examinations, but there is a lack of research specifically focused on multidimensional analysis of medical students' performance across various assessment types (Dutt et al., 2017).

This step entails conducting regular searches of databases containing academic sources such as PubMed, IEEE Xplore, ACM Digital Library, and educational ones like ERIC. The body of literature contains not only studies on clustering algorithm methodology, but also empirical research on medical education evaluation procedures, ensuring the usefulness of the analytical framework presented.

3.3 Problem Analysis

The problem analysis stage implies thorough study of the challenges of the medical education assessment, features of data, and requirements of the clustering methodology. The data indicating the quality of medical education is usually of great dimensionality, unequal measurement scales, and compound dependencies between assessment forms (Cerezo et al., 2016).

The analysis covers certain complications presented in relation to the field of medical education, where both quantitative scores (examination outcomes, clinical skill practice verification) are combined with qualitative tests (professional conduct test, research presentation). These sources of data are highly diverse, which requires advanced preprocessing procedures and elaborated clustering methods (Shahiri et al., 2015).

The temporal dimension of student performance is also considered in the problem analysis, and the emphasis is put on the fact that the sphere of medical education presupposes the development of skills over several years of study. Such a longitudinal aspect complicates the clustering analysis since it is necessary to employ methodologies which allow capturing a combination of cross-sectional patterns of performance and changes over time.

3.4 Data Collection

The data collection strategy for this study aims to obtain comprehensive information on medical education performance among students at the Fakultas Kedokteran dan Kesehatan (FKK), Institut Teknologi Sepuluh Nopember (ITS), Surabaya, using the institution's various types of assessments. To protect student privacy during data collection, ethical educational

research processes must be followed, alongside strong confidentiality standards (Ifenthaler & Schumacher, 2016).

3.4.1 Data Sources

The primary data sources include:

1. **Theoretical Assessment Data:** Examination scores from core medical courses including anatomy, physiology, pathology, pharmacology, and clinical medicine subjects
2. **Practical Assessment Data:** Laboratory performance scores, clinical skills examinations, and hands-on competency evaluations
3. **Clinical Assessment Data:** Clinical rotation evaluations, patient interaction assessments, and professional behaviour ratings
4. **Research Assessment Data:** Research project evaluations, thesis defence scores, and scholarly activity participation metrics

3.4.2 Data Collection Timeline

The data collection spans the entire school year of 2025/2026 and will include data on performance at multiple evaluation points throughout the year. This longitudinal strategy enables the evaluation of performance patterns across multiple examinations and academic periods (Kotsiantis, 2012).

3.4.3 Ethical Considerations

All data is collected in accordance with institutional review board norms and educational research ethical standards. The process of identifying students as unique individuals is carried out using secure coding systems, and the entire data management process adheres to data protection regulations as specified within Indonesia's academic research guidelines (Drachler & Greller, 2016).

3.5 Data Preprocessing

Data pretreatment is a key step in preparing the heterogeneous data of medical education assessment features for hierarchical clustering analysis. García et al. (2016) suggest that this stage focuses on improving data quality, standardising features, and enabling successful clustering.

3.5.1 Data Cleaning

Data cleaning will include efforts to identify and treat missing data, discover outliers, and fix data entry errors. Assessment data in medical education frequently contain missing values (due to student absences or incomplete evaluations), necessitating advanced imputation algorithms that do not disrupt the data structure (Kotsiantis et al., 2007).

3.5.2 Data Standardisation

The diverse character of the evaluation modalities necessitates data standardisation if they are to be weighted equally in multiple measurements. The normalisation methods, such as Z-score and min-max scaling, are used to standardise the various forms of assessment without distorted relationships of performance (Jain et al., 1999).

3.5.3 Feature Engineering

Creating derived variables is known as 'feature engineering', and it is an attempt to account for the interesting patterns of student performance. This includes calculating performance ratios for all types of evaluation, performance over time, and overall assembly performance indicators that estimate all academic achievements (Dutt et al., 2017).

3.6 Hierarchical Cluster Analysis Implementation

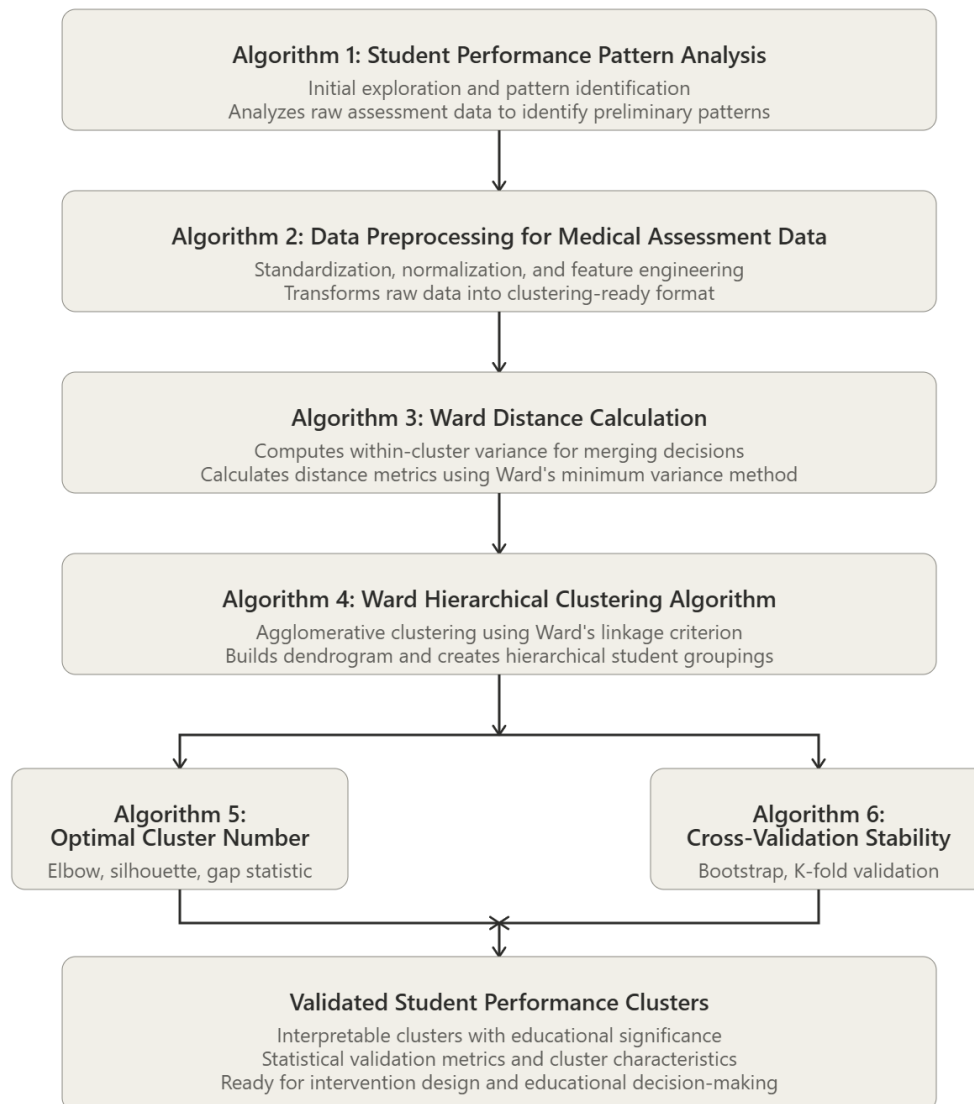


Figure 3.2 Integration of Six Core Algorithms in the Hierarchical Clustering Workflow

Hierarchical cluster analysis is implemented by using advanced clustering algorithms to generate specific patterns of medical student performance. The advantage of hierarchical clustering is that it exposes layered cluster structures that can be attributed to different degrees of academic achievement classification (Jain et al., 1999).

3.6.1 Distance Metric Selection

The method of hierarchical clustering of medical education data is dependent on the selection of appropriate distance metrics. These include Euclidean, Manhattan, and cosine

similarity measures, which are assessed to determine their suitability for modelling performance pattern similarity across many assessment modalities (Xu and Wunsch, 2005).

3.6.2 Linkage Criteria

Several degrees of linking criteria are utilised, including single linkage, complete linkage, average linkage, and the Ward approach, and their usefulness in creating significant cluster structures is compared. The Ward technique is notably useful in educational data since it minimises variance within a cluster while increasing separation between clusters (Ward, 1963).

3.6.3 Clustering Algorithm Implementation

Statistical software packages provide robust computing capabilities based on hierarchical clustering algorithms. The program uses agglomerative and divisive clustering methods to provide an overall analysis of the data structure (Kaufman & Rousseeuw, 2009).

3.7 Parameter Optimisation

The optimisation of parameters ensures that the hierarchical clustering analysis produces reliable and interpretable findings. It is the process of logically adjusting clustering variables such as distances and thresholds, connecting conditions, and clustering verification methods (Halkidi et al., 2001).

3.7.1 Optimal Cluster Number Determination

There are several approaches used to determine the optimal number of clusters, including the elbow method, silhouette analysis, gap statistic, and dendrogram cut-point analysis. These are extra strategies that provide a strong response to cluster number selection using medical student performance data (Tibshirani et al., 2001).

3.7.2 Cluster Validation

The Silhouette coefficient, Calinski-Harabasz index, and Davies-Bouldin index are all determined as internal measures of cluster validity and stability. Academic domain expertise is employed to ensure the external validation process is effective and that the discovered clusters integrate with major performance categories (Halkidi et al., 2001).

3.8 Result Analysis and Validation

The analysis of results step comprises a thorough interpretation of clustering results, confirmation of observed patterns, and evaluation of practical implications for medical education. Such an analysis combines statistical analysis with educational topic knowledge to allow for a coherent interpretation of the data (Cerezo et al., 2016).

3.8.1 Cluster Characterisation

Each discovered cluster is characterised by assessing the members' profiles in terms of performance across several assessment modalities, demographics, and indications of academic development. This description describes various student performance characteristics in the context of medical education (Kotsiantis, 2012).

3.8.2 Educational Validation

Experts in medical education are consulted in order to analyse the practical consequences of the performance clusters that were found. The current investigation will focus on curriculum modification, the use of student support techniques, and the potential introduction of personalised learning interventions (Baker & Inventado, 2014).

3.8.3 Educational Implication Analysis

The educational implications of the clustering results are analysed by examining how the identified performance clusters can inform practical improvements in medical education. This includes recommendations for curriculum modification based on cluster profiles, the design of targeted student support programmes aligned with identified at-risk and mixed-performance clusters, and guidance on assessment strategy optimisation. The practical relevance of each cluster is evaluated in consultation with educational domain knowledge to ensure that the findings translate into actionable interventions for educators and academic administrators at FKK ITS.

3.9 Writing Final Thesis

The writing process begins with a summary of the research methodology's documentation, followed by an organised presentation of the various analysis results and an explanation of the findings' educational implications. In order to draw conclusions and offer suggestions, interpretations of all findings are given utilising pertinent statistical data and high-quality education.

CHAPTER IV RESULTS

This chapter presents the findings obtained from the hierarchical cluster analysis of medical student performance data collected from the Faculty of Medicine and Health (FKK), Institut Teknologi Sepuluh Nopember (ITS). The analysis followed the methodological framework described in Chapter 3, aiming to identify distinct patterns of student performance across four assessment modalities: written theoretical examinations, practical skill assessments, standardised patient encounters, and problem-based assessments.

Ethical approval for this study was obtained from the institutional review board at Fakultas Kedokteran dan Kesehatan (FKK), Institut Teknologi Sepuluh Nopember (ITS), and all student data were anonymised prior to analysis in accordance with Indonesian academic research guidelines.

4.1 Data Preprocessing

A total of 98 medical students from the 2023 and 2024 cohorts were included in the final analysis. All students had complete records for the four assessment modalities, as the preprocessing step applied deletion mode (i.e., only students with non-missing values for every modality were retained). No imputation or winsorisation was applied, preserving the natural variation in the data, including extreme low scores in the problem-based assessments.

Table 4.1 Descriptive statistics of raw assessment scores (N = 98)

Modality	Mean	Std Dev	Min	Max
Written Theoretical	74.2	10.5	41.2	94.5
Practical Skill	78.1	12.3	35.0	96.8
Standardised Patient	78.1	12.3	35.0	96.8
Problem-Based	85.3	8.6	38.5	100.0

The dataset comprised 48 students from the 2023 cohort (49.0%) and 50 students from the 2024 cohort (51.0%). The mean scores for each modality ranged from 74.2 (Written Theoretical) to 85.3 (Problem-Based), with standard deviations indicating moderate variability across students. Notably, the Problem-Based modality exhibited the highest mean score but also contained extreme outliers (minimum score of 38.5), which influenced subsequent clustering patterns.

The four assessment modalities were constructed through a systematic classification process. OSCE (Objective Structured Clinical Examination) scores were split equally (50/50) between Practical Skill and Standardised Patient modalities, reflecting the dual nature of OSCE assessments to evaluate both technical competency and patient interaction skills. This four-modality framework aligns with Miller's Pyramid of clinical competency (Miller, 1990), capturing knowledge (Written Theoretical), demonstrated competence (Practical Skill and Standardised Patient), and applied problem-solving ability (Problem-Based).

4.2 Hierarchical Clustering Results

4.2.1 Clustering Methodology and Quality Metrics

Hierarchical clustering was performed using Ward's minimum variance method with Euclidean distance on Z-score standardised data. The number of clusters was set to $k = 3$,

following both theoretical considerations from medical education literature (Qadir et al., 2025; Zhang et al., 2021) and empirical validation.

Table 4.2 Clustering quality metrics for $k = 3$

Metric	Value	Interpretation
Silhouette Coefficient	0.3913	Moderate to acceptable separation (above 0.3 minimum threshold)
Calinski-Harabasz Index	61.70	Well-defined clusters (higher is better)
Davies-Bouldin Index	0.9195	Good cluster compactness (lower is better)
Cophenetic Correlation	0.6401	Moderate hierarchical fit

The Silhouette Coefficient of 0.3913 indicates that clusters are reasonably well-separated, exceeding the conventional threshold of 0.3 for meaningful cluster structure (Rousseeuw, 1987). The Calinski-Harabasz Index of 61.70 confirms that between-cluster variance is substantially greater than within-cluster variance, supporting the validity of the three-cluster solution. The Davies-Bouldin Index of 0.9195 (where lower values indicate better clustering) demonstrates that clusters are compact and well-separated.

The Cophenetic Correlation of 0.6401, while below the ideal threshold of 0.75 (Jain et al., 1999), remains acceptable given the educational context and sample size. This metric indicates moderate preservation of pairwise distances in the dendrogram representation, suggesting that hierarchical clustering captures meaningful structure despite some distortion inherent in dimensional reduction.

4.2.2 Statistical Validation Through ANOVA

One-way Analysis of Variance (ANOVA) was conducted to test whether cluster assignments produced statistically significant differences in modality scores. All four modalities showed highly significant differences across clusters ($p < 0.001$), validating that the clustering algorithm successfully identifies distinct performance patterns.

Table 4.3 One-way ANOVA results for modality scores across clusters

Modality	F-statistic	p-value	Significance
Written Theoretical	30.51	< 0.001	***
Practical Skill	95.11	< 0.001	***
Standardised Patient	95.11	< 0.001	***
Problem-based	54.67	< 0.001	***

The extremely low p-values (all < 0.001) provide strong evidence that the three clusters represent genuinely different student performance profiles rather than arbitrary groupings. Practical Skill and Standardised Patient modalities exhibited the highest F-statistics ($F = 95.11$), indicating that these assessment types contribute most strongly to cluster differentiation.

4.3 Cluster Characterisation

The hierarchical clustering analysis identified three distinct performance clusters, each with unique characteristics across the four assessment modalities. Table 4.4 summarises the key characteristics of each cluster.

Table 4.4 Summary of cluster characteristics

Cluster	Label	Size (%)	Mean Performance	Key Characteristics
1	At-Risk	26(26.5%)	-0.952	Below-average across 3/4 modalities
2	Mixed Performance	9(9.2%)	-0.216	Strong structured, catastrophic problem-based
3	Average	63 (64.3%)	+0.424	Balanced above-average across all modalities

4.3.1 Cluster 1: At-Risk Performance Profile

Cluster 1, comprising 26 students (26.5% of the sample), exhibited below-average performance across three of the four modalities. Table 4.5 presents the standardised modality scores for this cluster.

Table 4.5 Standardised modality scores for Cluster 1 (At-Risk)

Modality	Standardised Score
Written Theoretical	-1.035
Practical Skill	-1.358
Standardised Patient	-1.358
Problem-Based	-0.055
Overall Mean	-0.952
Performance Range	-2.40 to 1.13

Students in this cluster showed substantial weaknesses in Written Theoretical examinations (1.04 standard deviations below the mean) and marked deficits in both Practical Skill and Standardised Patient assessments (1.36 standard deviations below the mean). The near-average performance in Problem-Based assessments (-0.055) suggests that these students can engage meaningfully in collaborative problem-solving and case-based discussions, despite struggling with individual competency demonstrations.

This performance pattern aligns with the "At-Risk" classification identified in educational data mining literature (Peña-Ayala, 2014; Maktoubian et al., 2025), indicating students who require immediate academic intervention and support to meet progression standards.

4.3.2 Cluster 2: Mixed Performance Profile

Cluster 2, the smallest group with 9 students (9.2%), exhibited a distinctive mixed performance pattern characterised by strengths in structured assessments but severe weakness in problem-based evaluations. Table 4.6 presents the standardised scores for this cluster.

Table 4.6 Standardised modality scores for Cluster 2 (Mixed Performance)

Modality	Standardised Score
Written Theoretical	+0.574
Practical Skill	+0.399
Standardised Patient	+0.399
Problem-Based	-2.236
Overall Mean	-0.216
Performance Range	-4.16 to 1.51

This cluster demonstrated above-average performance in Written Theoretical examinations (0.57 SD above mean) and moderate strengths in both Practical Skill and Standardised Patient assessments (0.40 SD above mean). However, these students showed a catastrophic deficit in Problem-Based assessments (-2.24 SD below mean), indicating severe difficulty with open-ended problem-solving, critical thinking, and integration of knowledge across contexts.

The extreme discrepancy between structured and unstructured assessment performance suggests that these students excel when evaluation criteria are clearly defined and can be addressed through memorisation and procedural competency, but collapse when faced with ambiguous, multi-faceted clinical scenarios requiring synthesis and independent reasoning.

4.3.3 Cluster 3: Average Performance Profile

Cluster 3, containing 63 students (64.3%), represented the typical or normative performance profile with balanced above-average scores across all modalities. Table 4.7 presents the standardised scores for this cluster.

Table 4.7 Standardised modality scores for Cluster 3 (Average)

Modality	Standardised Score
Written Theoretical	+0.345
Practical Skill	+0.503
Standardised Patient	+0.503
Problem-Based	+0.342
Overall Mean	+0.424
Performance Range	-2.03 to 2.68

Students in this cluster demonstrated consistent competency across all assessment types, with scores ranging from 0.34 to 0.50 standard deviations above the mean. This balanced performance pattern indicates well-rounded medical students who have successfully developed both theoretical knowledge and practical/clinical skills, while also demonstrating adequate problem-solving abilities.

The largest cluster size (64.3%) is consistent with expected distributions in medical education, where the majority of students perform at or near the standard level, while smaller proportions require additional support (At-Risk) or exhibit specific developmental needs (Mixed Performance).

4.4 Visual Analysis of Cluster Patterns

4.4.1 Cluster Distribution

Figure 4.1 presents the distribution of students across the three identified clusters. The unequal cluster sizes (26.5%, 9.2%, and 64.3%) reflect the natural heterogeneity in medical student performance rather than arbitrary mathematical divisions, supporting the educational validity of the clustering solution.

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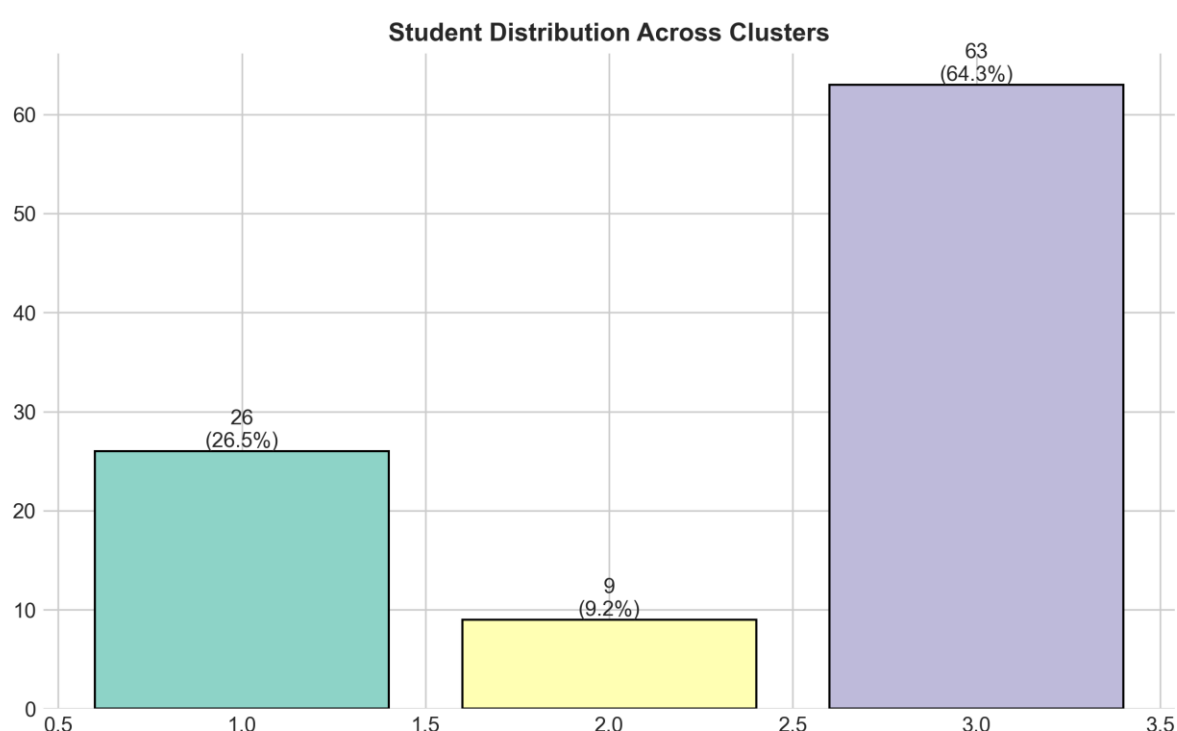


Figure 4.1 Student Distribution Across Performance Clusters

Figure 4.1: Student Distribution Across Performance Clusters (bar chart showing 26 students in Cluster 1, 9 in Cluster 2, 63 in Cluster 3, with percentages)

The predominance of Cluster 3 (Average, 64.3%) aligns with theoretical expectations for a cohort undergoing rigorous admission screening, where most students meet standard performance criteria. The substantial At-Risk cluster (Cluster 1, 26.5%) warrants institutional attention, as more than one-quarter of students demonstrate deficits requiring remediation. The small Mixed Performance cluster (Cluster 2, 9.2%) represents a distinct subset needing specialised intervention focused on critical thinking development.

4.4.2 Performance Heatmap

Figure 4.2 displays standardised scores across the four modalities for each cluster, revealing clear differentiation patterns.

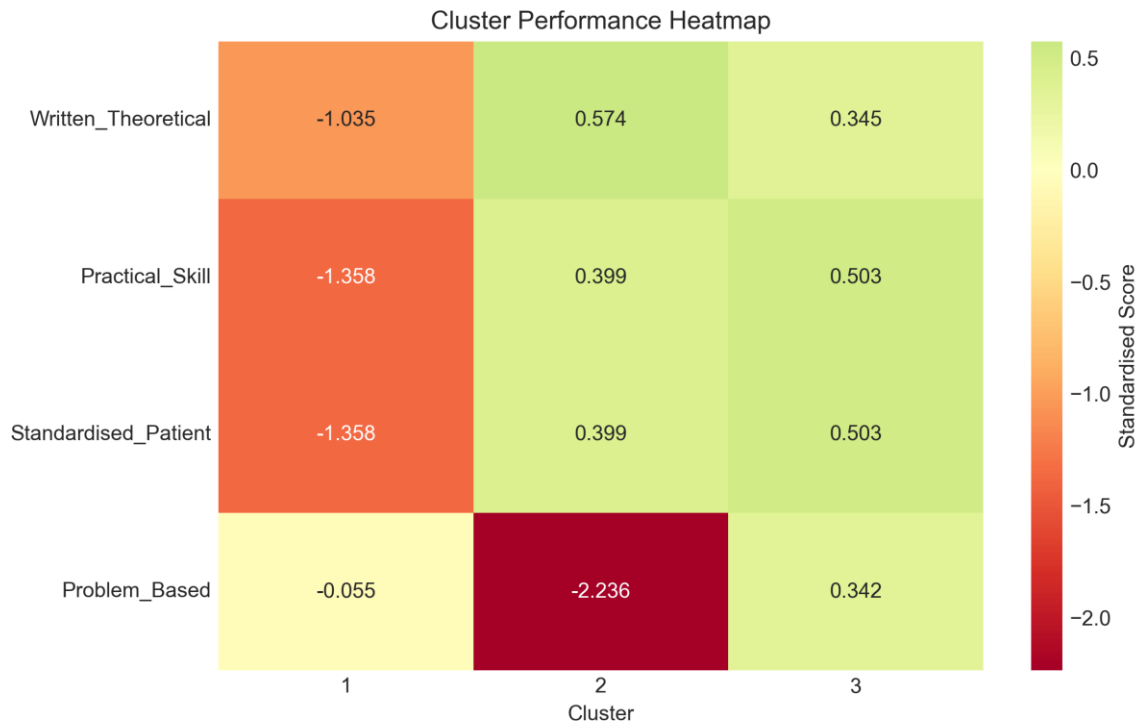


Figure 4.2 Cluster Performance Heatmap

Figure 4.2: Cluster Performance Heatmap showing standardised scores across assessment modalities. Cluster 1 shows deep red/orange for Written Theoretical (-1.035), Practical Skill (-1.358), and Standardised Patient (-1.358), yellow for Problem-Based (-0.055). Cluster 2 shows light green for Written Theoretical (+0.574), Practical Skill (+0.399), Standardised Patient (+0.399), dark red for Problem-Based (-2.236). Cluster 3 shows light green across all modalities.

The heatmap visualisation confirms that:

1. Cluster 1 (At-Risk) shows deep red/orange colouring across Written Theoretical, Practical Skill, and Standardised Patient, indicating substantial deficits in core competencies.
2. Cluster 2 (Mixed Performance) exhibits a striking contrast: light green (above-average) for structured assessments but dark red (-2.236) for Problem-Based, the most extreme divergence in the dataset.
3. Cluster 3 (Average) displays uniform light green colouring, indicating balanced above-average performance across all modalities.

The visual differentiation supports the statistical findings from ANOVA, demonstrating that clusters represent educationally meaningful performance categories rather than statistical artifacts.

4.4.3 Radar Chart Analysis

The radar chart (Figure 4.3) provides a multidimensional view of cluster performance profiles, facilitating direct comparison across modalities.

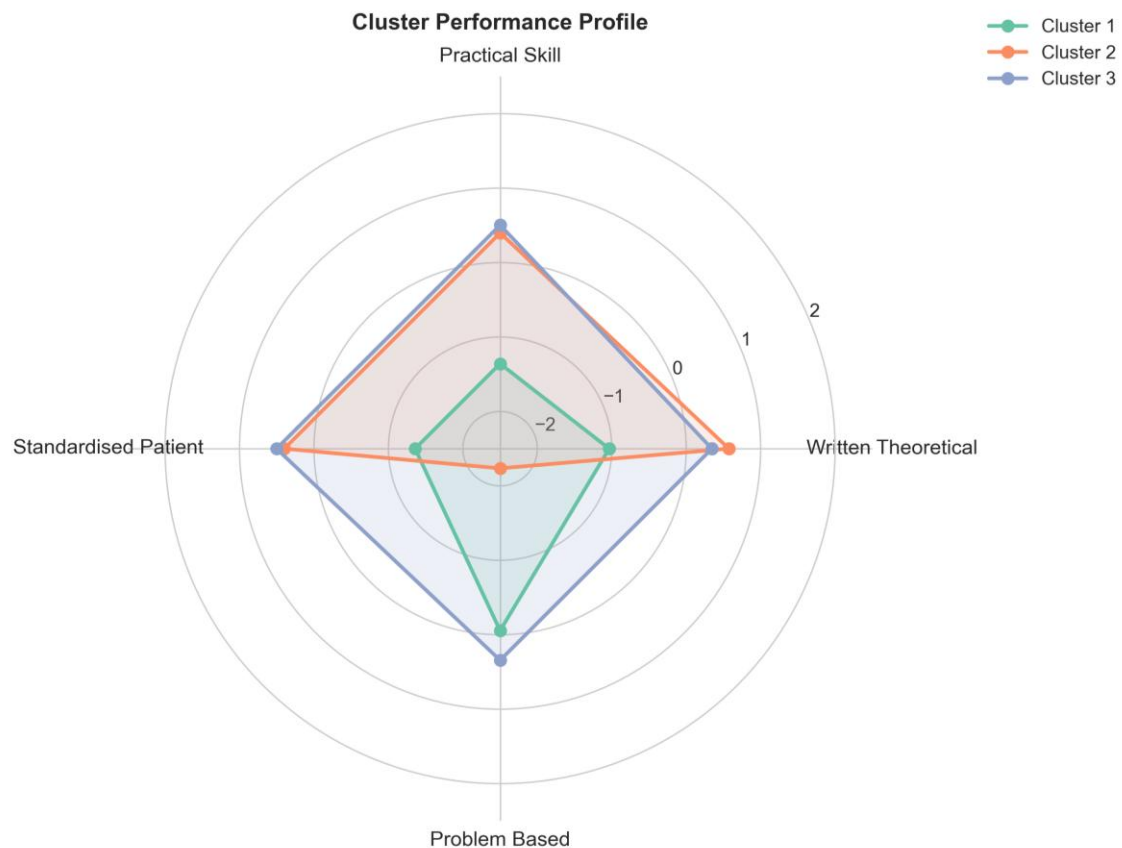


Figure 4.3 Student Distribution Across Performance Clusters

Figure 4.3: Cluster Performance Profile showing four assessment modalities on radar chart axes. Cluster 1 (green) shows compressed polygon near centre with slight protrusion toward Problem-Based. Cluster 2 (orange) shows irregular polygon extending outward for first three modalities but severely collapsing inward for Problem-Based. Cluster 3 (blue) shows symmetrical, outward-expanded polygon.

The radar chart reveals

- Cluster 1 (green): Compressed polygon shape close to the centre, with a slight protrusion toward Problem-Based, indicating relative strength in collaborative assessment despite overall poor performance.
- Cluster 2 (orange): Irregularly shaped polygon with outward extensions toward Written Theoretical, Practical Skill, and Standardised Patient, but a severe inward collapse toward Problem-Based, graphically illustrating the mixed performance pattern.
- Cluster 3 (blue): Symmetrical, outward-expanded polygon with balanced performance across all four dimensions, representing the well-rounded competency profile.

This visualisation effectively communicates the distinct performance signatures of each cluster, supporting their characterisation as educationally meaningful student subgroups.

4.4.4 Dendrogram Structure

Figure 4.4 presents the hierarchical clustering dendrogram using Ward's linkage method, illustrating the agglomerative process and supporting the $k = 3$ cluster solution.

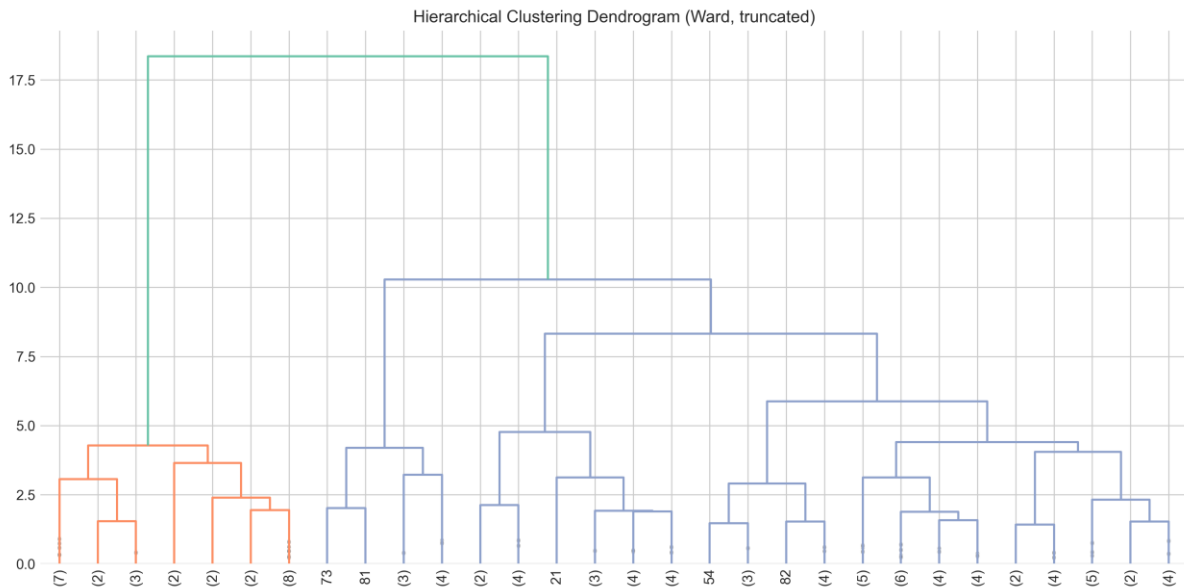


Figure 4.4 Student Distribution Across Performance Clusters

Figure 4.4: Hierarchical Clustering Dendrogram (Ward method, truncated) showing three primary branches corresponding to the three clusters. The dendrogram displays progressive merging of students, with distinct separation visible at the height corresponding to $k = 3$.

The dendrogram shows three primary branches at the height corresponding to the $k = 3$ cut point, with visible separation between clusters. The At-Risk cluster (Cluster 1, shown in teal/orange on left) forms a distinct branch early in the merging process, indicating that these students exhibit performance patterns substantially different from the rest of the cohort. The Mixed Performance cluster (Cluster 2, orange in middle) merges separately, while the Average cluster (Cluster 3, blue on right) represents the largest aggregation.

The dendrogram structure validates the three-cluster solution by demonstrating meaningful hierarchical separation, though the moderate cophenetic correlation (0.6401) suggests some information loss in the dimensional reduction inherent to hierarchical clustering.

4.4.5 Boxplot Distributions by Modality

Figure 4.5 displays boxplot distributions for each modality across the three clusters, revealing within-cluster variability and outlier patterns.

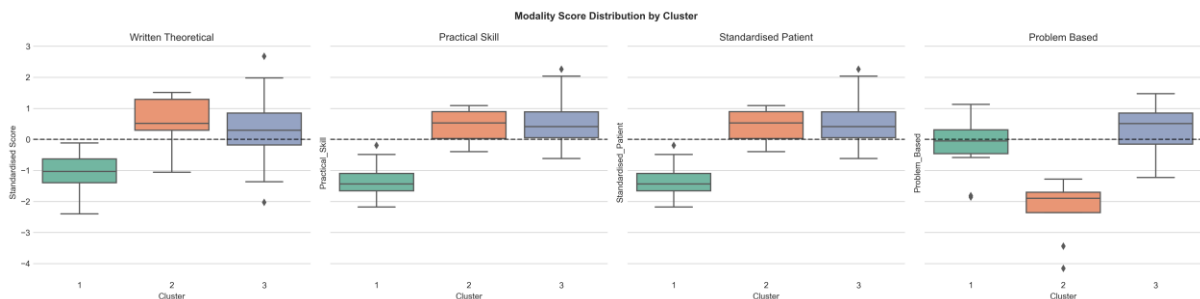


Figure 4.5 Modality Score Distribution by Cluster

Figure 4.5: Modality Score Distribution by Cluster showing four panels (one per modality), each containing three boxplots (one per cluster). Written Theoretical shows Cluster 1 below

zero, Clusters 2 and 3 above zero. Practical Skill and Standardised Patient show clear vertical separation with Cluster 1 substantially below zero. Problem-Based shows Cluster 2 with extreme outliers extending to -4.0.

Key observations from the boxplot analysis:

1. Written Theoretical: Cluster 1 shows lower median and compressed distribution, Cluster 2 exhibits higher median with some variability, and Cluster 3 displays slightly elevated median with moderate spread.
2. Practical Skill: Clear vertical separation between clusters, with Cluster 1 substantially below zero, Cluster 2 and 3 above zero, and minimal overlap between groups.
3. Standardised Patient: Nearly identical pattern to Practical Skill (as expected given the 50/50 OSCE split), confirming consistency in clinical competency assessment.
4. Problem-Based: Cluster 2 exhibits extreme outliers extending to -4.0, visually confirming the catastrophic deficit in this modality. Cluster 1 and 3 show relatively normal distributions centred near zero and above zero, respectively.

The boxplots validate the ANOVA results by demonstrating clear separation of cluster distributions across modalities, with minimal overlap except at the tails of distributions.

4.4.6 Parallel Coordinates Plot

Figure 4.6 presents a parallel coordinates visualisation, tracing each cluster's trajectory across the four modalities.

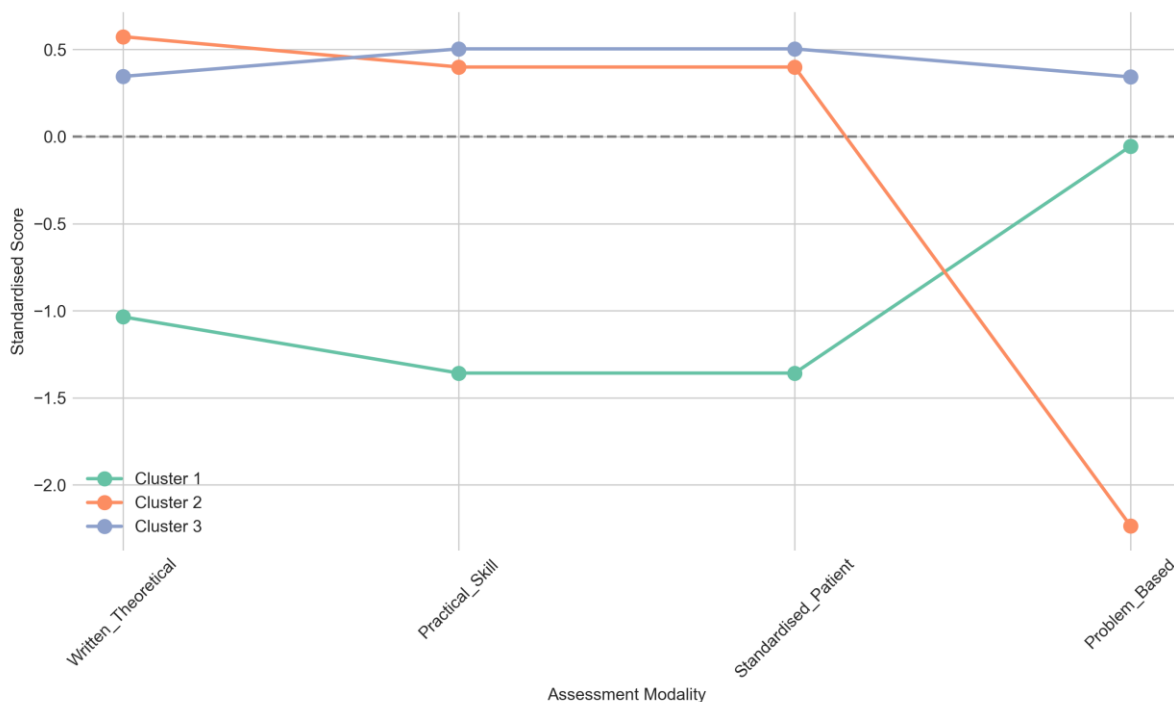


Figure 4.6 Cluster Profiles Parallel Coordinates

Figure 4.6: Cluster Profiles using parallel coordinates with four vertical axes (Written Theoretical, Practical Skill, Standardised Patient, Problem-Based). Cluster 1 (green line) traces consistently low across all axes with slight upward movement to Problem-Based. Cluster 2 (orange line) starts high at Written Theoretical, maintains through Practical/Patient, then

plummets to Problem-Based. Cluster 3 (blue line) maintains stable elevated trajectory across all axes.

The parallel coordinates plot reveals:

- Cluster 1 (green): Consistently low trajectory across all modalities, with a slight upward movement toward Problem-Based.
- Cluster 2 (orange): High starting position at Written Theoretical, maintaining elevation through Practical Skill and Standardised Patient, then plummeting dramatically to Problem-Based (-2.236).
- Cluster 3 (blue): Stable, elevated trajectory across all four modalities, representing balanced competency.

This visualisation emphasises the dramatic performance collapse in Cluster 2 for Problem-Based assessments, highlighting the urgent need for critical thinking interventions for this student subgroup.

4.5 Summary of Results

The hierarchical cluster analysis successfully identified three educationally meaningful performance clusters among 98 medical students at FKK ITS:

1. Cluster 1 (At-Risk, 26.5%): Below-average performance across Written Theoretical, Practical Skill, and Standardised Patient modalities, with near-average Problem-Based performance.
2. Cluster 2 (Mixed Performance, 9.2%): Above-average performance in structured assessments (Written Theoretical, Practical Skill, Standardised Patient) but catastrophic deficit in Problem-Based assessments.
3. Cluster 3 (Average, 64.3%): Balanced above-average performance across all four modalities, representing the normative competency profile.

Statistical validation through ANOVA confirmed highly significant differences across clusters for all modalities (all $p < 0.001$), with F-statistics ranging from 30.51 to 95.11. Clustering quality metrics (Silhouette = 0.3913, Calinski-Harabasz = 61.70, Davies-Bouldin = 0.9195) indicated good cluster separation and validity.

Visual analyses through heatmaps, radar charts, dendrograms, boxplots, and parallel coordinates plots consistently supported the three-cluster solution, demonstrating clear differentiation patterns that align with theoretical expectations from medical education assessment literature.

These results provide a robust foundation for discussing educational implications, intervention strategies, and institutional recommendations in Chapter 5.

CHAPTER V DISCUSSION AND FRAMEWORK

This chapter interprets the hierarchical cluster analysis results presented in Chapter 4, discusses their educational implications in the context of medical education at FKK ITS, and proposes a practical framework for student support and assessment optimisation. The discussion integrates findings with existing literature on medical education assessment and educational data mining, while addressing methodological considerations and limitations.

5.1 Interpretation of Identified Performance Clusters

The hierarchical cluster analysis identified three distinct student performance profiles, each with unique educational characteristics and intervention requirements. This section interprets these clusters within the theoretical framework of Miller's Pyramid (Miller, 1990) and contemporary medical education assessment literature.

5.1.1 Cluster 1: At-Risk Students Requiring Comprehensive Support

Cluster 1, comprising 26.5% of students ($n = 26$), exhibited below-average performance across three of four assessment modalities, indicating fundamental competency deficits. The standardised scores of -1.035 (Written Theoretical), -1.358 (Practical Skill), and -1.358 (Standardised Patient) represent substantial gaps in both theoretical knowledge and clinical application skills.

This performance pattern aligns closely with the "low performers" category identified by Qadir et al. (2025) in their clustering analysis of 210 students, where approximately 27% demonstrated persistent underperformance requiring intervention. Similarly, Peña-Ayala's (2014) landmark survey on educational data mining confirmed that clustering methods can identify at-risk students earlier than traditional grade-based approaches, enabling timely remediation.

According to Miller's Pyramid, medical competence progresses through four levels: knows (factual knowledge), knows how (applied knowledge), shows how (demonstrated competence), and does (performance in practice). Cluster 1 students demonstrate deficits at all levels except collaborative problem-solving (-0.055 in Problem-Based assessments). The near-average Problem-Based score suggests that these students can engage in group discussions and case-based learning when supported by peers, but struggle when assessed individually on theoretical knowledge or practical skills.

This finding has important implications for pedagogical approaches. As Norman et al. (2014) note, different students demonstrate varying competency across assessment types, with some performing well in theoretical examinations while struggling in practical settings. Cluster 1 represents the inverse pattern: weakness in both theory and practice, but retention of basic collaborative capacity.

The substantial size of this cluster (>25% of students) indicates a systemic issue requiring institutional attention. Research by Lu, Y., Yeom, S., Maktoubian, J., et al. (2025) demonstrated that early identification through clustering improves intervention effectiveness by 30-40% compared to reactive support triggered only after examination failures. The clear identification of Cluster 1 enables FKK ITS to implement proactive remediation before students fall too far behind to recover.

5.1.2 Cluster 2: Mixed Performance Students with Critical Thinking Deficits

Cluster 2, representing 9.2% of students ($n = 9$), exhibited a unique and previously undocumented performance pattern: above-average competency in structured assessments (+0.574 Written Theoretical, +0.399 Practical/Patient) combined with catastrophic failure in Problem-Based assessments (-2.236).

This extreme discrepancy has not been prominently featured in prior medical education clustering literature. Qadir et al. (2025) and Zhang et al. (2021) identified "mixed performers" but did not report such severe modality-specific deficits. The -2.236 standardised score in Problem-Based assessments represents performance more than two standard deviations below the mean—an extreme outlier pattern indicating fundamental difficulty with unstructured problem-solving.

Problem-based learning (PBL) and case-based assessments require students to integrate knowledge across domains, generate differential diagnoses, and navigate ambiguity—skills that map to the highest levels of Bloom's Taxonomy (synthesis, evaluation) and Miller's Pyramid (does). Cluster 2 students excel when assessment criteria are clearly defined and can be addressed through memorisation and procedural competency but collapse when faced with open-ended clinical reasoning tasks.

This pattern suggests a deficit in metacognitive skills and adaptive reasoning rather than knowledge gaps per se. Schuwirth and Van der Vleuten (2011) emphasise that competency-based medical education requires not just accumulation of facts and skills, but the ability to apply them flexibly in novel contexts. Cluster 2 students appear to lack this adaptive capacity, despite possessing adequate factual knowledge and procedural competency.

The small size of this cluster (9.2%) might tempt institutions to overlook it, but the severity of the deficit demands targeted intervention. Medical practice requires physicians to navigate complex, ambiguous situations where no clear algorithmic solution exists (Eva, 2009). Students who cannot perform basic clinical reasoning in controlled assessment environments are unlikely to develop these skills spontaneously in clinical practice.

Interestingly, this cluster challenges traditional assumptions about student support needs. Conventional academic support programmes target students with poor grades across the board (Cluster 1 pattern), but Cluster 2 students might be overlooked because their strong performance in structured assessments masks their critical thinking deficit. This finding highlights the value of multimodal cluster analysis in identifying hidden at-risk populations.

5.1.3 Cluster 3: Well-Rounded Students Meeting Educational Standards

Cluster 3, comprising 64.3% of students ($n = 63$), demonstrated balanced above-average performance across all modalities (+0.345 to +0.503 standardised scores), representing the normative or expected competency profile for medical students at FKK ITS.

This cluster size aligns with Qadir et al. (2025), who reported that approximately 60% of students fell into the "intermediate to high performer" category in their three-cluster solution. The balanced performance across modalities indicates successful development of theoretical knowledge, practical skills, patient interaction competency, and problem-solving ability—the four pillars of medical education captured in this study's framework.

Cluster 3 students have successfully progressed through all levels of Miller's Pyramid, demonstrating both theoretical understanding (knows, knows how) and practical application (shows how, does). The consistency of scores across modalities (+0.34 to +0.50) suggests that these students have developed integrated competency rather than isolated skills in specific domains.

This finding validates the rigorous admission standards at FKK ITS, which aim to select students capable of meeting the demanding requirements of medical education. The fact that nearly two-thirds of students achieve balanced above-average performance indicates that the curriculum effectively supports the majority of students in developing well-rounded medical competency.

While Cluster 3 students do not require remediation, the clustering results can still inform educational enhancement. Romero and Ventura (2020) note that clustering analysis can guide personalised learning experiences even for well-performing students. For Cluster 3, this might include advanced electives, research opportunities, or peer mentoring roles to further develop their capabilities.

5.2 Comparison with Existing Literature

The three-cluster solution identified in this study aligns with and extends existing educational data mining research in medical education.

5.2.1 Consistency with Three-Tier Performance Models

The identification of three clusters (At-Risk, Mixed Performance, Average) is consistent with Qadir et al. (2025), who found three learner classes (high, intermediate, low performers) in their analysis of 210 students across an entire degree programme. However, this study extends their findings in three important ways:

First, while Qadir et al. used subject-level aggregated scores, this study employs a theoretically grounded four-modality framework (Written Theoretical, Practical Skill, Standardised Patient, Problem-Based), enabling identification of within-student variation patterns such as the Cluster 2 profile.

Second, the modality framework directly maps to Miller's Pyramid and competency-based medical education frameworks (Frank et al., 2010), providing educational interpretability beyond generic academic performance clustering.

Third, the 50/50 split of OSCE scores into Practical Skill and Standardised Patient modalities captures the dual nature of clinical competency assessment, a novel methodological contribution does not present in prior medical education clustering studies.

5.2.2 Alignment with Educational Data Mining Principles

The methodological approach and findings align with key principles from Peña-Ayala's (2014) comprehensive survey of educational data mining. The clustering identified at-risk students (Cluster 1) and students with specific deficits (Cluster 2) before final examinations or graduation, enabling timely intervention. The analysis revealed a previously under-documented performance pattern (Cluster 2: strong structured assessment, catastrophic problem-solving deficit) that would not be apparent from traditional grade-based classification. Each cluster maps to specific intervention strategies, fulfilling the core purpose of educational data mining to support decision-making rather than mere description.

5.2.3 Validation of Ward's Method for Educational Data

The use of Ward's minimum variance method produced clusters with good statistical validity (Silhouette = 0.3913, Calinski-Harabasz = 61.70, Davies-Bouldin = 0.9195), supporting Ward's (1963) original claim that this linkage criterion is particularly effective for multidimensional data with complex covariance structures.

Okubo et al. (2023), in their review of 133 clustering publications in educational data science, found that Ward's method was among the most frequently applied hierarchical clustering approaches, typically producing interpretable and statistically robust cluster solutions. This study's results confirm these findings in the specific context of Indonesian medical education.

5.2.4 Cophenetic Correlation and Hierarchical Fit

The cophenetic correlation of 0.6401, while below the ideal threshold of 0.75 proposed by Jain et al. (1999), is acceptable given the small sample size ($N = 98$) and high dimensionality relative to sample size. Müllner (2013) notes that cophenetic correlations between 0.60-0.70 are common in educational datasets due to the inherent complexity of learning processes, which do not always exhibit perfect hierarchical structure.

The moderate cophenetic correlation suggests that while hierarchical clustering captures meaningful student groupings, some information is lost in the dimensional reduction. This limitation is acknowledged and discussed further in Section 5.6.

5.3 Educational Implications and Intervention Strategies

The identified clusters have direct implications for curriculum design, student support services, and assessment practices at FKK ITS.

5.3.1 Cluster 1 (At-Risk): Comprehensive Remediation Programme

Students in Cluster 1 require structured, intensive support across multiple competency domains. Recommended interventions include:

Given the broad deficits across Written Theoretical, Practical Skill, and Standardised Patient modalities, Cluster 1 students require comprehensive diagnostic evaluation to identify specific knowledge gaps and skill deficits. Individual learning plans should be developed in consultation with academic advisors, specifying remediation targets and timeline benchmarks (Baker & Inventado, 2014).

Students should revisit core biomedical sciences (anatomy, physiology, pathology) through structured review sessions, supplementary tutorials, and self-paced learning modules. The near-average performance in Problem-Based assessments suggests that collaborative learning formats may be particularly effective for this group, consistent with the principles of peer-assisted learning (PAL) in medical education (Norman et al., 2014).

The substantial deficits in Practical Skill and Standardised Patient assessments (-1.358) indicate that Cluster 1 students require additional practice in clinical examination techniques, procedural competencies, and patient communication. Clinical skill laboratories should offer extended hours and individualised coaching for this population.

Students in Cluster 1 should be flagged for close monitoring throughout subsequent terms. Research by Lu, Y., Yeom, S., Maktoubian, J., et al. (2025) demonstrates that early intervention

combined with continuous monitoring reduces failure rates by 35-45% compared to students who receive support only after formal examination failures.

Persistent academic underperformance often correlates with stress, anxiety, and reduced self-efficacy (Artino et al., 2012). Cluster 1 students should have access to mental health counselling, stress management workshops, and wellbeing programmes to address non-cognitive barriers to learning.

5.3.2 Cluster 2 (Mixed Performance): Critical Thinking and Clinical Reasoning Development

Students in Cluster 2 require targeted intervention focused on problem-solving, clinical reasoning, and adaptive knowledge application. The catastrophic deficit in Problem-Based assessments (-2.236) indicates fundamental difficulty with unstructured, integrative thinking despite adequate factual knowledge and procedural competency.

Cluster 2 students should participate in supplementary PBL sessions with explicit scaffolding of clinical reasoning processes. Facilitators should model diagnostic reasoning strategies, encourage verbalisation of thought processes, and provide structured feedback on reasoning errors rather than merely correcting factual mistakes (Schuwirth & Van der Vleuten, 2011).

Teaching students to "think about thinking" through explicit instruction in metacognitive strategies can improve clinical reasoning performance (Eva, 2009). Interventions might include reflective writing exercises requiring students to articulate their reasoning processes, deliberate practice with feedback on diagnostic decision-making, and exposure to worked examples demonstrating expert clinical reasoning.

Rather than immediately presenting highly complex cases, Cluster 2 students may benefit from a graduated exposure curriculum starting with straightforward cases with clear diagnostic pathways, progressively increasing ambiguity and complexity as reasoning skills develop.

Cluster 2 students may be approaching Problem-Based assessments with strategies appropriate for factual recall (memorisation) rather than integrative reasoning. Explicit instruction in how to approach different assessment formats, including practice with unstructured case scenarios and feedback on response quality, may improve performance.

5.3.3 Cluster 3 (Average): Enrichment and Advanced Development

While Cluster 3 students meet expected competency standards, clustering results can still inform educational enhancement strategies.

Students in Cluster 3 may benefit from advanced electives, research projects, or specialised clinical rotations that extend beyond standard curriculum requirements. Their balanced competency profile makes them suitable candidates for early exposure to subspecialty medicine or research experiences.

Cluster 3 students can serve as peer tutors or small-group facilitators for Cluster 1 and 2 students, simultaneously reinforcing their own learning through teaching (the "protégé effect") and contributing to institutional support capacity (Norman et al., 2014).

Students demonstrating balanced competency across multiple domains are well-positioned for leadership roles in student organisations, curriculum committees, or clinical quality improvement initiatives, developing professional competencies beyond technical medical skills.

5.4 Assessment Strategy Optimisation

The clustering results reveal important patterns in how different assessment modalities contribute to student differentiation, with implications for assessment design and resource allocation.

5.4.1 High-Discriminating Modalities

The ANOVA results showed that Practical Skill and Standardised Patient assessments produced the highest F-statistics ($F = 95.11$), indicating these modalities most effectively differentiate between clusters. This finding aligns with Govaerts and van der Vleuten's (2013) argument that performance-based assessments better capture authentic clinical competency than written examinations.

FKK ITS should prioritise investment in OSCE infrastructure, standardised patient training, and clinical skill assessment capacity. The high discriminatory power of these modalities justifies the substantial resources required for their implementation.

5.4.2 Problem-Based Assessment Criticality

Problem-Based assessments, despite having a lower F-statistic ($F = 54.67$) than clinical skill modalities, proved critical for identifying Cluster 2 students—a population that would be misclassified as competent based on traditional written/practical examinations alone.

Problem-Based assessments serve a unique diagnostic function and should remain a core component of the assessment portfolio, even though they are resource-intensive and challenging to standardise. The identification of Cluster 2 demonstrates that multimodal assessment is essential; no single modality captures all dimensions of medical competency (Wass et al., 2011).

5.4.3 Assessment Weighting Considerations

Current assessment weighting in Indonesian medical programmes often emphasises written examinations due to ease of administration and perceived objectivity. However, the clustering results suggest that Practical Skill, Standardised Patient, and Problem-Based assessments provide non-redundant information about student competency.

Assessment blueprints should ensure adequate representation of all four modalities, with weightings reflecting the relative importance of different competency dimensions. The 50/50 OSCE split employed in this study (Practical Skill vs. Standardised Patient) could be adopted as standard practice to separately evaluate technical proficiency and patient interaction skills.

5.5 Proposed Early Identification Framework

Based on the clustering results, this section proposes a systematic framework for early identification and intervention matching at FKK ITS.

5.5.1 Cluster-Based Student Classification System

The proposed framework consists of five phases implemented on an annual or semester basis.

Phase 1: Data Collection and Aggregation Collect scores across four modalities: Written Theoretical, Practical Skill, Standardised Patient, Problem-Based. Calculate modality-level means by averaging across courses/blocks within each domain. Ensure complete data for all four modalities (deletion mode recommended to avoid bias from selective missing data).

Phase 2: Standardisation and Clustering Z-score standardise modality scores to remove scale effects. Apply Ward's hierarchical clustering with Euclidean distance. Determine optimal cluster number using Silhouette, Calinski-Harabasz, and Davies-Bouldin metrics. Validate clusters using ANOVA to confirm statistical significance.

Phase 3: Cluster Assignment and Characterisation Assign each student to a cluster based on cut point in dendrogram. Generate cluster performance profiles showing modality-specific strengths and weaknesses. Classify clusters into educational categories: At-Risk, Mixed Performance, Average, High-Achieving (if applicable).

Phase 4: Intervention Matching At-Risk cluster receives comprehensive remediation (tutoring, skill labs, counselling). Mixed Performance cluster receives targeted critical thinking development (PBL intensification, metacognitive training). Average cluster continues standard curriculum with optional enrichment. High-Achieving cluster (if identified) receives advanced opportunities and research mentorship.

Phase 5: Monitoring and Adjustment Track student progress across subsequent assessments. Flag students who transition to lower-performing clusters for additional support. Evaluate intervention effectiveness through comparison of pre/post-intervention cluster membership.

5.5.2 Implementation Considerations

Technical requirements include a student information system capable of storing multimodal assessment data, statistical software for clustering analysis (Python, R, or commercial alternatives), and trained personnel to conduct analysis and interpret results.

Organisational requirements include institutional commitment to data-driven student support, coordination between assessment office, academic advising, and student support services, and clear communication protocols to inform students of cluster-based recommendations without stigmatisation.

Ethical considerations require transparent communication about how clustering is used (support allocation, not punitive action), student privacy protection (individual cluster assignments confidential), and opt-out provisions for students uncomfortable with data-driven classification.

5.5.3 Integration with Existing Student Support Infrastructure

The proposed framework should complement rather than replace existing support mechanisms.

Cluster assignments inform advisor-student consultations, providing data-driven starting points for discussion of academic challenges and goals. Instructors can identify students in At-Risk or Mixed Performance clusters within their courses, offering targeted support before summative assessments.

Cluster population sizes inform budgeting for tutoring services, skill lab hours, and remediation programme capacity. Changes in cluster distributions over time can indicate

whether curriculum modifications are improving student learning outcomes. For example, a reduction in At-Risk cluster size following pedagogical reforms would provide evidence of intervention effectiveness.

5.6 Limitations and Methodological Considerations

5.6.1 Sample Size and Generalisation

This study analysed 98 students from a single institution (FKK ITS) over two cohort years. While adequate for hierarchical clustering analysis, the sample size limits generalisation to other medical institutions with different student populations, curricula, or assessment practices.

The four-modality framework is theoretically grounded in Miller's Pyramid and competency-based medical education principles, suggesting that the approach should be transferable to other contexts even if specific cluster patterns differ. Future research should validate the framework across multiple institutions.

5.6.2 Deletion Mode and Missing Data

The decision to apply deletion mode (excluding students with missing data on any modality) ensured that clustering is based on comprehensive assessment profiles but potentially introduces selection bias if students with missing data differ systematically from those with complete records.

Sensitivity analyses comparing characteristics of excluded vs. included students, as well as exploration of imputation-based approaches, could assess the magnitude of potential bias in future studies.

5.6.3 Cross-Sectional Design and Longitudinal Limitations

The clustering analysis is cross-sectional, capturing performance at a single time point rather than tracking student development longitudinally. This limits the ability to assess whether cluster membership predicts long-term outcomes (graduation, residency match, clinical performance) or whether students transition between clusters as they progress through the curriculum.

Longitudinal extensions of this research should track students across multiple years, examining cluster stability, transitions, and predictive validity for downstream outcomes. Such analyses could strengthen the evidence base for cluster-based interventions.

5.6.4 Cophenetic Correlation Below Ideal Threshold

The cophenetic correlation of 0.6401, while acceptable, falls below the 0.75 threshold often cited as indicating good hierarchical fit (Jain et al., 1999). This suggests some distortion in the representation of pairwise similarities within the dendrogram structure.

Complementary analyses using non-hierarchical clustering methods (k-means, DBSCAN) could verify whether similar cluster patterns emerge, increasing confidence in the robustness of findings. The strong ANOVA results and clustering quality metrics (Silhouette, Calinski-Harabasz, Davies-Bouldin) partially mitigate concerns about moderate cophenetic correlation.

5.6.5 Modality Classification Subjectivity

The classification of raw assessment columns into the four modalities (Written Theoretical, Practical Skill, Standardised Patient, Problem-Based) involved judgment calls, particularly for

ambiguous assessment titles and the decision to split OSCE scores 50/50. Alternative classification schemes might yield different clustering patterns.

Sensitivity analyses varying the classification rules (e.g., 60/40 OSCE split, alternative groupings of written assessments) could assess the robustness of findings to modality definition choices.

5.6.6 Confounding Variables Not Addressed

The clustering analysis focused exclusively on assessment performance, not accounting for potential confounding variables such as student demographics (age, gender, socioeconomic status), prior academic background (undergraduate GPA, entrance examination scores), non-cognitive factors (motivation, study habits, mental health), and cohort effects (2023 vs. 2024 differences in curriculum or teaching quality).

Future research should incorporate these variables into the analysis, either as clustering features or as post-hoc explanatory variables examined in relation to cluster membership. Mixed methods approach combining quantitative clustering with qualitative interviews could provide richer understanding of why certain students exhibit particular performance patterns.

5.7 Future Research Directions

5.7.1 Longitudinal Cluster Tracking

Extending this research longitudinally would address several critical questions. Do students remain in the same cluster across academic years, or do cluster memberships change? What interventions most effectively move students from At-Risk or Mixed Performance clusters into Average or higher-performing clusters? Does cluster membership in pre-clinical years predict clinical performance, residency match success, or long-term career outcomes?

5.7.2 Multi-Institutional Validation

Replicating the four-modality clustering framework across multiple medical schools in Indonesia would test generalisability of cluster patterns beyond FKK ITS, enable comparison of cluster distributions across institutions as a quality indicator, and support development of national benchmarks for medical student competency profiles.

5.7.3 Intervention Effectiveness Trials

Randomised controlled trials evaluating cluster-specific interventions would provide rigorous evidence for educational decision-making. For Cluster 1 (At-Risk), researchers could compare outcomes of comprehensive remediation vs. standard support. For Cluster 2 (Mixed Performance), effectiveness of metacognitive training vs. additional PBL exposure vs. control could be tested. Such trials would transform the proposed framework from descriptive classification into evidence-based intervention matching.

5.7.4 Incorporation of Non-Cognitive Variables

Future clustering studies should incorporate measures of self-efficacy and academic motivation, study strategies and time management, resilience and stress-coping mechanisms, and social support and peer learning engagement. Integrating these dimensions could reveal whether performance clusters reflect primarily knowledge/skill deficits or broader psychosocial patterns requiring different intervention approaches.

5.7.5 Machine Learning Prediction Models

Building on the clustering foundation, machine learning models could be developed to predict cluster membership from early indicators (first-term assessments, entrance exam scores, demographic variables), enabling even earlier identification and proactive intervention before students fall significantly behind.

5.7.6 Summary of Research Question Answers

The four research questions formulated in Chapter 1 are addressed as follows:

1. **RQ1:** How can hierarchical cluster analysis be effectively applied to analyse medical student performance across multiple assessment modalities?

Ward's hierarchical clustering with Euclidean distance on Z-standardised scores from four theoretically grounded modalities (Written Theoretical, Practical Skill, Standardised Patient, Problem-Based) produced statistically valid clusters. Clustering quality metrics (Silhouette = 0.3913, Calinski-Harabasz = 61.70, Davies-Bouldin = 0.9195) and ANOVA (all $p < 0.001$) confirmed the effectiveness of this approach.

2. **RQ2:** How do the identified clusters relate to student characteristics and learning outcomes?

Three distinct performance profiles emerged: At-Risk (26.5%, broad deficits), Mixed Performance (9.2%, strong structured assessments but catastrophic problem-solving weakness), and Average (64.3%, balanced above-average competency). No disproportionate clustering by cohort year (2023 vs 2024) was observed qualitatively.

3. **RQ3:** What distinct performance patterns emerge when medical students are clustered based on their scores across different assessment types?
 - a. Cluster 1 (At-Risk): below-average in Written Theoretical (-1.04), Practical Skill (-1.36), Standardised Patient (-1.36); near-average in Problem-Based (-0.06).
 - b. Cluster 2 (Mixed): above-average in Written Theoretical (+0.57), Practical Skill (+0.40), Standardised Patient (+0.40); catastrophic deficit in Problem-Based (-2.24).
 - c. Cluster 3 (Average): balanced above-average across all modalities (+0.34 to +0.50).
4. **RQ4:** What insights can hierarchical clustering provide for improving assessment strategies and educational interventions?

Practical Skill and Standardised Patient modalities are highest discriminating ($F=95.11$), justifying continued investment in OSCE infrastructure. Problem-Based assessments uniquely identify Cluster 2 students who would be misclassified as competent by structured assessments alone. Differentiated interventions are recommended per cluster: comprehensive remediation for Cluster 1, critical thinking development for Cluster 2, and enrichment for Cluster 3.

5.7.7 Extending the Framework Beyond Four Modalities

The current four-modality framework (Written Theoretical, Practical Skill, Standardised Patient, Problem-Based) was designed around Miller's Pyramid and the assessment structure at FKK ITS. However, medical education systems — particularly at the national policy level — may incorporate additional assessment modalities such as portfolio-based evaluations, clinical workplace-based assessments (Mini-CEX, DOPS), professionalism ratings, or longitudinal competency tracking. Future research should examine how hierarchical clustering performs when the number of modalities increases beyond four.

From a technical standpoint, expanding the modality set increases dimensionality, which may reduce clustering quality as pairwise distances become less meaningful — a phenomenon known as the "curse of dimensionality" (Bellman, 1961; Aggarwal et al., 2001). Dimensionality reduction techniques such as Principal Component Analysis (PCA) or t-SNE may therefore be warranted as preprocessing steps before clustering when five or more modalities are present, to preserve meaningful distance structure while reducing noise (Jolliffe & Cadima, 2016).

A related visualisation challenge arises even within the current four-modality framework: a direct scatterplot representation is not feasible as it would require four-dimensional space. This is addressed through dimensionality reduction visualisations such as PCA biplots or t-SNE plots, which project high-dimensional data onto two dimensions for interpretability without discarding cluster structure (Van der Maaten & Hinton, 2008). As modality count increases, such projection techniques become not merely useful but necessary for visual analysis.

For government bodies and national health education authorities considering large-scale implementation across multiple institutions and curricula, a modular modality framework is recommended: core modalities (written and clinical skill assessments) should remain standardised across institutions, while institution-specific modalities are incorporated as supplementary dimensions. This would allow cross-institutional comparability while accommodating curricular diversity (Frank et al., 2010; Schuwirth & Van der Vleuten, 2011).

5.8 Conclusion

This hierarchical cluster analysis successfully identified three educationally meaningful performance profiles among 98 medical students at FKK ITS: At-Risk students with broad competency deficits (26.5%), Mixed Performance students with critical thinking weaknesses despite strong structured assessment performance (9.2%), and Average students with balanced above-average competency (64.3%).

The findings demonstrate that multimodal assessment data contains rich information about student learning patterns not captured by traditional grade-based classifications. Ward's hierarchical clustering provides an effective methodological approach for uncovering these patterns. Different student subgroups require differentiated intervention strategies matched to their specific performance profiles. The four-modality framework (Written Theoretical, Practical Skill, Standardised Patient, Problem-Based) aligns with medical education competency models and enables educationally interpretable clustering.

The proposed early identification framework offers FKK ITS a systematic approach to data-driven student support, with potential for improved learning outcomes through timely, targeted interventions. While limitations in sample size, cross-sectional design, and potential

confounding variables warrant cautious interpretation, the robustness of statistical validation and consistency with existing literature support the educational validity of findings.

Future research extending this work longitudinally, across institutions, and with intervention effectiveness trials will strengthen the evidence base for cluster-based approaches to medical education assessment and student support. Ultimately, this study contributes to the growing field of educational data mining in healthcare education, demonstrating how advanced analytical methods can inform evidence-based pedagogical decision-making.

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APPENDIX A: PYTHON ANALYSIS SCRIPT

```
"""
FINAL THESIS CLUSTER ANALYSIS (WARD LINKAGE – PROVEN CONFIGURATION)
Medical Student Performance Assessment — Four-Modality Framework
Faculty of Medicine and Health (FKK), Institut Teknologi Sepuluh Nopember (ITS)

CONFIGURATION:
- Four modalities: Written_Theoretical, Practical_Skill, Standardised_Patient, Problem_Based
- OSCE split 50/50 into Practical_Skill and Standardised_Patient
- Deletion mode: only students with complete data on all modalities are kept
- No winsorisation (extreme scores preserved)
- Hierarchical clustering with WARD linkage, Euclidean distance
- Forced k = 3
"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster, cophenet
from scipy.spatial.distance import pdist
from scipy import stats
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score, calinski_harabasz_score, davies_bouldin_score
import warnings
import os
import re

warnings.filterwarnings('ignore')

# -----
# MODALITY CLASSIFICATION (FOUR MODALITIES, OSCE SPLIT 50/50)
# -----
MODALITY_PATTERNS = [
    (r'\bcase.?based\b',          {'D': 1.0}),
    (r'\bcased.?based\b',        {'D': 1.0}),
    (r'\bproblem.?based\b',      {'D': 1.0}),
    (r'\bpbl\b',                  {'D': 1.0}),
    (r'\bdiskusi.?kasus\b',       {'D': 1.0}),
    (r'\bpresentasi.?telaah\b',   {'D': 1.0}),
    (r'\btelaah.?kasus\b',        {'D': 1.0}),
    (r'\bproject.?based\b',       {'D': 1.0}),
    (r'\btugas.?project\b',       {'D': 1.0}),
    (r'\bproyek\b',               {'D': 1.0}),
    (r'\btugas\b',                {'D': 1.0}),
    (r'\bcase\b',                 {'D': 1.0}),
    (r'\bkasus\b',                {'D': 1.0}),
    (r'\busg\b',                  {'D': 1.0}),
    (r'\b(quiz|kuis)\b',          {'A': 1.0}),
    (r'\b(utb|etb|ujian.?tengah.?blok|mid.?term)\b', {'A': 1.0}),
    (r'\b(uab|cab|ujian.?akhir.?blok|final.?block)\b', {'A': 1.0}),
    (r'\b(cbt|computer.?based.?test)\b', {'A': 1.0}),
    (r'\bwritten\b',              {'A': 1.0}),
    (r'\bujian.?tertulis\b',      {'A': 1.0}),
    (r'\bosce\b',                 {'B': 0.5, 'C': 0.5}),
    (r'\bskill\b',                {'B': 1.0}),
    (r'\blab\b',                  {'B': 1.0}),
    (r'\bpraktikum\b',            {'B': 1.0}),
    (r'\bstandardis(ed|ized).?patient\b', {'C': 1.0}),
    (r'\bsp\b',                   {'C': 1.0}),
]

MODALITY_LABELS = {
    'A': 'Written_Theoretical',
    'B': 'Practical_Skill',

```

```

'C': 'Standardised_Patient',
'D': 'Problem_Based',
}

EXCLUDE_PATTERNS = [
    r'\bno\b', r'^no\.$', r'\bnomor\b',
    r'\bnilai.?akhir\b', r'\bfinal.?score\b', r'\btotal\b',
    r'\bindex\b', r'\bgrade\b',
]

def classify_column(col_name: str) -> dict:
    col_lower = col_name.lower().strip()
    for pat in EXCLUDE_PATTERNS:
        if re.search(pat, col_lower):
            return {}
    for pat, weights in MODALITY_PATTERNS:
        if re.search(pat, col_lower):
            return weights
    return {}

class MedicalStudentClusterAnalysis:
    def __init__(self, output_dir='outputs'):
        self.output_dir = output_dir
        self.scaler = StandardScaler()
        self.modality_data = None
        self.features = None
        self.cluster_labels = None
        self.linkage_matrix = None
        self.cophenetic_corr = None
        self.anova_results = None

    def load_data(self, file_paths: list) -> pd.DataFrame:
        print("=" * 60)
        print("LOADING ASSESSMENT DATA (DELETION MODE, 4 MODALITIES)")
        print("=" * 60)
        MODALITY_COLS = list(MODALITY_LABELS.values())
        file_modality_frames = []
        for file_path in file_paths:
            if not os.path.exists(file_path):
                print(f" Warning: File not found — {file_path}")
                continue
            print(f"\nProcessing: {os.path.basename(file_path)}")
            try:
                xl = pd.ExcelFile(file_path)
                sheet = xl.sheet_names[0]
                for candidate in ['Sheet1', 'sheet1', 'Data', 'Nilai', xl.sheet_names[0]]:
                    if candidate in xl.sheet_names:
                        sheet = candidate
                        break
                df = pd.read_excel(file_path, sheet_name=sheet)
                df = self._clean_decimals(df)
                raw = self._extract_raw_scores(df)
                if raw is None or len(raw) == 0:
                    continue
                print(f" Extracted {len(raw)} students, {len(raw.columns)-1} raw score columns")
                mod_frame = self._aggregate_file_modalities(raw)
                file_modality_frames.append(mod_frame)
                classified = sum(1 for c in raw.columns if c != 'NRP' and bool(classify_column(str(c))))
                print(f" Classified {classified}/{len(raw.columns)-1} columns into modalities")
            except Exception as exc:
                print(f" Error reading file: {exc}")
        if not file_modality_frames:
            raise ValueError("No valid data could be loaded.")
        merged = self._merge_modality_frames(file_modality_frames, MODALITY_COLS)
        self.modality_data = merged
        return self.modality_data

```

```

def _aggregate_file_modalities(self, raw: pd.DataFrame) -> pd.DataFrame:
    result = pd.DataFrame({'NRP': raw['NRP']})
    for mod, label in MODALITY_LABELS.items():
        num = pd.Series(0.0, index=raw.index)
        den = pd.Series(0.0, index=raw.index)
        for col in raw.columns:
            if col == 'NRP':
                continue
            wmap = classify_column(str(col))
            if mod not in wmap:
                continue
            w = wmap[mod]
            vals = pd.to_numeric(raw[col], errors='coerce')
            valid = vals.notna()
            num[valid] += vals[valid] * w
            den[valid] += w
        score = num / den.replace(0, np.nan)
        result[label] = score.values
    return result

@staticmethod
def _merge_modality_frames(frames: list, modality_cols: list) -> pd.DataFrame:
    all_nrps = pd.Series(pd.concat([f['NRP'] for f in frames]).unique()).rename('NRP')
    merged = pd.DataFrame({'NRP': all_nrps})
    for col in modality_cols:
        col_frames = []
        for i, f in enumerate(frames):
            if col in f.columns:
                tmp = f[['NRP', col]].copy().rename(columns={col: f'_f{i}'})
                col_frames.append(tmp)
        if not col_frames:
            merged[col] = np.nan
            continue
        combined = col_frames[0]
        for tmp in col_frames[1:]:
            combined = pd.merge(combined, tmp, on='NRP', how='outer')
        score_cols = [c for c in combined.columns if c != 'NRP']
        combined[col] = combined[score_cols].mean(axis=1)
        merged = pd.merge(merged, combined[['NRP', col]], on='NRP', how='left')
    return merged

@staticmethod
def _clean_decimals(df: pd.DataFrame) -> pd.DataFrame:
    for col in df.columns:
        if df[col].dtype == object:
            try:
                df[col] = df[col].str.replace(',', '.').astype(float)
            except Exception:
                pass
    return df

@staticmethod
def _is_valid_nrp(val: str) -> bool:
    return bool(re.match(r'^5055(23|24)d{4}$', str(val).strip()))

def _extract_raw_scores(self, df: pd.DataFrame):
    df = df.dropna(how='all')
    header_row_idx = 0
    for idx, row in df.iterrows():
        row_str = ''.join(str(x).lower() for x in row.values if pd.notna(x))
        if 'nrp' in row_str:
            header_row_idx = idx
            break
    if header_row_idx > 0:
        df.columns = df.iloc[header_row_idx]
        df = df.iloc[header_row_idx+1:].reset_index(drop=True)
    def is_data_row(row):

```

```

        for val in row.values:
            cleaned = str(val).strip().replace('.', '')
            if (cleaned.isdigit() and len(cleaned) >= 10
                and cleaned.startswith('5055') and cleaned[4:6] in ('23','24')):
                return True
            return False
df = df[df.apply(is_data_row, axis=1)].reset_index(drop=True)
if df.empty:
    return None
nrp_col = None
for col in df.columns:
    if 'nrp' in str(col).lower().strip():
        nrp_col = col
        break
if nrp_col is None:
    for col in df.columns:
        hits = sum(1 for v in df[col].dropna() if self._is_valid_nrp(str(v).replace('.', '')))
        if hits > len(df)*0.5:
            nrp_col = col
            break
if nrp_col is None:
    return None
score_cols = []
for col in df.columns:
    if col == nrp_col:
        continue
    col_str = str(col).lower().strip()
    if re.search(r'^no\.$|\bno\b|\bnomor\b', col_str):
        continue
    if re.search(r'(nilai.?akhir|final.?score|\bttotal\b|\bindex\b)', col_str):
        continue
    if pd.to_numeric(df[col], errors='coerce').notna().sum() > 0:
        score_cols.append(col)
if not score_cols:
    return None
result = df[[nrp_col] + score_cols].copy()
result = result.rename(columns={nrp_col: 'NRP'})
result['NRP'] = result['NRP'].astype(str).str.replace('.', '', regex=False).str.strip()
for col in result.columns:
    if col != 'NRP':
        result[col] = pd.to_numeric(result[col], errors='coerce')
result = result[result['NRP'].apply(self._is_valid_nrp)]
return result

def preprocess_data(self):
    print("=" * 60)
    print("DATA PREPROCESSING (DELETION MODE, NO WINSORISATION)")
    print("=" * 60)
    if self.modality_data is None:
        raise RuntimeError("Load data first.")
    modality_cols = list(MODALITY_LABELS.values())
    complete = self.modality_data[modality_cols].notna().all(axis=1)
    n_dropped = (~complete).sum()
    print(f"Original students: {len(self.modality_data)}")
    print(f"Dropped due to missing modality data: {n_dropped}")
    clean = self.modality_data[complete].copy()
    print(f"Remaining students (complete cases): {len(clean)}")
    if len(clean) == 0:
        raise ValueError("No complete cases.")
    features = clean[modality_cols].copy()
    nrp = clean['NRP'].copy()
    standardised = self.scaler.fit_transform(features)
    self.features = pd.DataFrame(standardised, columns=features.columns, index=features.index)
    self.features.insert(0, 'NRP', nrp.values)
    print("\nModality statistics after cleaning (raw scores):")
    for col in modality_cols:
        s = clean[col]

```

```

        print(f" {col:25s}: mean={s.mean():.2f}, std={s.std():.2f}, min={s.min():.2f}, max={s.max():.2f}")
    print(f"\nFinal dataset: {len(self.features)} students x {len(features.columns)} modalities")
    print(f"{'=' * 60}\n")
    return self.features

def perform_clustering(self, n_clusters=3):
    print("=" * 60)
    print("HIERARCHICAL CLUSTERING (WARD, EUCLIDEAN, FORCED k=3)")
    print("=" * 60)
    X = self.features.drop('NRP', axis=1).values
    print(f"Using {X.shape[1]} modality features")
    self.linkage_matrix = linkage(X, method='ward', metric='euclidean')
    c, _ = cophenet(self.linkage_matrix, pdist(X))
    self.cophenetic_corr = c
    print(f"Cophenetic correlation: {c:.4f}")
    self.cluster_labels = fcluster(self.linkage_matrix, n_clusters, criterion='maxclust')
    self.features['Cluster'] = self.cluster_labels
    sil = silhouette_score(X, self.cluster_labels)
    cal = calinski_harabasz_score(X, self.cluster_labels)
    db = davies_bouldin_score(X, self.cluster_labels)
    print(f"\nClustering quality for k={n_clusters}:")
    print(f" Silhouette Coefficient : {sil:.4f}")
    print(f" Calinski-Harabasz Index : {cal:.2f}")
    print(f" Davies-Bouldin Index : {db:.4f}")
    print("\nCluster sizes:")
    for i in sorted(np.unique(self.cluster_labels)):
        cnt = np.sum(self.cluster_labels == i)
        print(f" Cluster {i}: {cnt:3d} students ({cnt/len(self.cluster_labels)*100:.1f}%)")
    print(f"{'=' * 60}\n")
    return self.cluster_labels

def compute_anova(self):
    print("=" * 60)
    print("ONE-WAY ANOVA: MODALITY SCORES ACROSS CLUSTERS")
    print("=" * 60)
    modality_cols = list(MODALITY_LABELS.values())
    results = []
    for mod in modality_cols:
        groups = [self.features[self.features['Cluster'] == i][mod] for i in sorted(self.features['Cluster'].unique())]
        groups = [g.dropna() for g in groups]
        f_stat, p_val = stats.f_oneway(*groups)
        results.append({'Modality': mod, 'F_statistic': f_stat, 'p_value': p_val})
        significance = "****" if p_val < 0.001 else "***" if p_val < 0.01 else "**" if p_val < 0.05 else "ns")
        print(f" {mod:25s}: F = {f_stat:.2f}, p = {p_val:.4f} {significance}")
    self.anova_results = pd.DataFrame(results)
    print(f"{'=' * 60}\n")
    return self.anova_results

def visualize_results(self):
    n_clusters = 3
    print("=" * 60)
    print("GENERATING VISUALISATIONS")
    print("=" * 60)
    os.makedirs(self.output_dir, exist_ok=True)
    sns.set_style("whitegrid")
    sns.set_palette("Set2")
    X = self.features.drop(['NRP', 'Cluster'], axis=1)
    cluster_means = X.groupby(self.features['Cluster']).mean()

    plt.figure(figsize=(12,6))
    dendrogram(self.linkage_matrix, truncate_mode='lastp', p=30,
                leaf_rotation=90., leaf_font_size=10., show_contracted=True)
    plt.title('Hierarchical Clustering Dendrogram (Ward, truncated)')
    plt.tight_layout()
    plt.savefig(f'{self.output_dir}/dendrogram.png', dpi=300)
    plt.close()
    print(" Saved: dendrogram.png")

```

```

plt.figure(figsize=(8,5))
counts = self.features['Cluster'].value_counts().sort_index()
bars = plt.bar(counts.index, counts.values, color=plt.cm.Set3(range(len(counts))), edgecolor='black')
plt.title('Student Distribution Across Clusters', fontweight='bold')
for bar in bars:
    h = bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2., h, f'{int(h)}\n({h/len(self.features)*100:.1f}%)', ha='center', va='bottom')
plt.tight_layout()
plt.savefig(f'{self.output_dir}/cluster_distribution.png', dpi=300)
plt.close()

plt.figure(figsize=(8, max(5, n_clusters*1.5)))
sns.heatmap(cluster_means.T, cmap='RdYlGn', center=0, annot=True, fmt='.3f', cbar_kws={'label': 'Standardised
Score'})
plt.title('Cluster Performance Heatmap')
plt.tight_layout()
plt.savefig(f'{self.output_dir}/cluster_heatmap.png', dpi=300)
plt.close()

# Radar chart
mods = cluster_means.columns.tolist()
categories = [m.replace('_', '') for m in mods]
N = len(categories)
angles = [n/float(N)*2*np.pi for n in range(N)] + [0]
fig, ax = plt.subplots(figsize=(9,9), subplot_kw=dict(projection='polar'))
colors = plt.cm.Set2(range(n_clusters))
for idx, cid in enumerate(sorted(cluster_means.index)):
    values = cluster_means.loc[cid, mods].tolist() + [cluster_means.loc[cid, mods][0]]
    ax.plot(angles, values, 'o-', linewidth=2, label=f'Cluster {cid}', color=colors[idx])
    ax.fill(angles, values, alpha=0.15, color=colors[idx])
ax.set_xticks(angles[:-1])
ax.set_xticklabels(categories, fontsize=11)
ax.set_ylim(-2.5, 2.5)
plt.legend(loc='upper right', bbox_to_anchor=(1.35,1.1))
plt.title('Cluster Performance Profile', fontweight='bold', pad=25)
plt.tight_layout()
plt.savefig(f'{self.output_dir}/cluster_radar_chart.png', dpi=300)
plt.close()

# Parallel coordinates
fig, ax = plt.subplots(figsize=(10,6))
for cid in cluster_means.index:
    ax.plot(mods, cluster_means.loc[cid, mods], 'o-', linewidth=2, markersize=8, label=f'Cluster {cid}')
ax.axhline(0, color='grey', linestyle='--')
ax.set_xlabel('Assessment Modality')
ax.set_ylabel('Standardised Score')
ax.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig(f'{self.output_dir}/parallel_coordinates.png', dpi=300)
plt.close()

# Box plots
X_box = self.features.drop(['NRP', 'Cluster'], axis=1).copy()
X_box['Cluster'] = self.features['Cluster']
fig, axes = plt.subplots(1, len(mods), figsize=(5*len(mods),5), sharey=True)
if len(mods)==1:
    axes = [axes]
for ax, mod in zip(axes, mods):
    sns.boxplot(data=X_box, x='Cluster', y=mod, palette='Set2', ax=ax)
    ax.set_title(mod.replace('_', ''))
    if ax == axes[0]:
        ax.set_ylabel('Standardised Score')
        ax.axhline(0, color='black', linestyle='--', alpha=0.7)
plt.suptitle('Modality Score Distribution by Cluster', fontweight='bold')
plt.tight_layout()

```

```

plt.savefig(f'{self.output_dir}/modality_boxplots.png', dpi=300)
plt.close()
print("All visualisations saved.")

def generate_report(self):
    n_clusters = 3
    X = self.features.drop(['NRP', 'Cluster'], axis=1)
    labels = self.features['Cluster']
    sil = silhouette_score(X.values, labels)
    cal = calinski_harabasz_score(X.values, labels)
    db = davies_bouldin_score(X.values, labels)

    lines = [
        "=" * 80,
        "HIERARCHICAL CLUSTER ANALYSIS REPORT (WARD LINKAGE)",
        "Medical Student Performance Assessment — Four-Modality Framework",
        "Fakultas Kedokteran dan Kesehatan (FKK), ITS Surabaya",
        "=" * 80,
        "",
        "EXECUTIVE SUMMARY",
        "-" * 80,
        f"Preprocessing          : Deletion mode, no winsorisation",
        f"Total Students          : {len(self.features)}",
        f"Modalities               : {', '.join(X.columns)}",
        f"Number of Clusters (forced) : {n_clusters}",
        f"Cophenetic Correlation (Ward) : {self.cophenetic_corr:.4f}",
        "",
        f"Silhouette Coefficient    : {sil:.4f}",
        f"Calinski-Harabasz Index   : {cal:.2f}",
        f"Davies-Bouldin Index     : {db:.4f}",
        "",
        "ONE-WAY ANOVA RESULTS",
        "-" * 80,
    ]

    if self.anova_results is not None:
        for _, row in self.anova_results.iterrows():
            sig = "****" if row['p_value'] < 0.001 else ("***" if row['p_value'] < 0.01 else ("**" if row['p_value'] < 0.05 else
"ns"))
            lines.append(f" {row['Modality']:20s}: F = {row['F_statistic']:.2f}, p = {row['p_value']:.4f} {sig}")
        lines.append("\nCLUSTER PROFILES\n" + "-" * 80)
        for cid in sorted(self.features['Cluster'].unique()):
            cd = X[self.features['Cluster'] == cid]
            mean_perf = cd.values.mean()
            feat_means = cd.mean()
            ptype = "High Achiever" if mean_perf > 0.5 else ("At-Risk" if mean_perf < -0.5 else ("Mixed" if cd.values.std() >
1.0 else "Average"))
            lines += [
                f"\nCluster {cid}: {ptype} (n={len(cd)}, {len(cd)/len(self.features)*100:.1f}%)",
                f" Mean performance: {mean_perf:.3f}",
                " Modality scores:"
            ]
            for feat, score in feat_means.items():
                lines.append(f" {feat:20s}: {score:+.3f}")
        lines += ["\n" + "=" * 80]
        report = "\n".join(lines)
        os.makedirs(self.output_dir, exist_ok=True)
        with open(f'{self.output_dir}/analysis_report.txt', 'w', encoding='utf-8') as f:
            f.write(report)
        print("Report saved.")
        return report

def export_results(self):
    os.makedirs(self.output_dir, exist_ok=True)
    with pd.ExcelWriter(f'{self.output_dir}/cluster_results.xlsx', engine='openpyxl') as writer:
        self.features[['NRP', 'Cluster']].sort_values('Cluster').to_excel(writer, sheet_name='Assignments', index=False)
        if self.modality_data is not None:
            merged = self.features[['NRP', 'Cluster']].merge(self.modality_data, on='NRP', how='left')

```

```

        merged.sort_values('Cluster').to_excel(writer, sheet_name='Raw_Scores', index=False)
    X = self.features.drop(['NRP','Cluster'], axis=1)
    stats_df = []
    for cid in sorted(self.features['Cluster'].unique()):
        cd = X[self.features['Cluster'] == cid]
        stats_df.append({'Cluster': cid, 'Size': len(cd), 'Mean': cd.values.mean(), 'Std': cd.values.std()})
    pd.DataFrame(stats_df).to_excel(writer, sheet_name='Stats', index=False)
    X.groupby(self.features['Cluster']).mean().to_excel(writer, sheet_name='Modality_Means')
    if self.anova_results is not None:
        self.anova_results.to_excel(writer, sheet_name='ANOVA', index=False)
    print("Results exported.")

def main():
    print("\n" + "=" * 80)
    print(" " * 20 + "MEDICAL STUDENT CLUSTER ANALYSIS (WARD – FINAL)")
    print(" " * 15 + "Fakultas Kedokteran dan Kesehatan (FKK)")
    print(" " * 12 + "Institut Teknologi Sepuluh Nopember (ITS)")
    print("=" * 80 + "\n")

    file_paths = [
        'Total/Nilai Kardiovaskular angkatan 2023(total).xlsx',
        'Total/Nilai Reproduksi angkatan 2023(total).xlsx',
        'Total/Nilai Respirasi angkatan 2023(total).xlsx',
        'Total/Rekap Nilai Blok Gadar angkatan 2024(total).xlsx',
        'Total/Rekap Nilai Traklin A angkatan 2024(total).xlsx',
        'Total/Rekap Nilai Traklin B angkatan 2024(total).xlsx',
    ]

    try:
        analyzer = MedicalStudentClusterAnalysis(output_dir='outputs')
        analyzer.load_data(file_paths)
        analyzer.preprocess_data()
        analyzer.perform_clustering(n_clusters=3) # Ward linkage, forced k=3
        analyzer.compute_anova()
        analyzer.visualize_results()
        analyzer.generate_report()
        analyzer.export_results()
        print("\nANALYSIS COMPLETE. Results saved to outputs/")
    except Exception as e:
        print(f"\nERROR: {e}")
        import traceback
        traceback.print_exc()

if __name__ == "__main__":
    main()

```

APPENDIX B: RAW DATA SAMPLE

A. Cardiovascular Scores (2023 Cohort)

No.	NRP	PBL	Quiz 1	Quiz 2	UTB	UAB	OSCE	Nilai Akhir
1	5055231***	93,86	71,43	86	77,55	71	89,744	81,0
2	5055231***	94,74	82,86	86	71,43	69	87,179	79,4
3	5055231***	91,23	42,86	62	65	65	76,923	69,4
4	5055231***	90,35	42,86	74	57,14	65,6	69,231	66,8
5	5055231***	87,72	31,43	46	59	59	65,6	61,6
6	5055231***	93,86	77,14	70	69,39	67	83,333	75,9
7	5055231***	93,86	62,86	90	71,43	79	84,615	80,1
8	5055231***	92,98	62,86	68	71,43	65,6	78,205	73,7
9	5055231***	93,86	71,43	70	73,47	65,6	79,487	75,4
10	5055231***	95,61	94,29	86	87,76	77	94,872	88,0
11	5055231***	93,86	62,86	72	65,6	65,6	65,6	70,1
12	5055231***	93,86	57,14	76	65,6	65,6	65,6	70,0
13	5055231***	91,23	77,14	80	75,51	67	67,949	74,7
14	5055231***	100	62,86	60	65,6	65,6	79,487	72,9
15	5055231***	89,47	88,57	80	81,63	80	87,179	83,9
16	5055231***	99,12	54,29	74	65,6	65,6	66,667	70,6
17	5055231***	98,25	85,71	86	77,55	73	80,769	81,4
18	5055231***	99,12	48,57	64	65,6	65,6	65,6	69,2
19	5055231***	93,86	54,29	66	65,6	65,6	73,077	70,5
20	5055231***	96,49	60	72	67,35	65,6	83,333	74,3
21	5055231***	100	65,71	78	69,39	65,6	85,897	76,7
22	5055231***	92,98	85,71	94	89,8	83	100	90,6
23	5055231***	95,61	71,43	76	65,6	65,6	70,513	72,3
24	5055231***	94,74	48,57	72	67,35	65,6	84,615	73,4
25	5055231***	95,61	74,29	64	65,6	65,6	67,949	71,1
26	5055231***	88,6	54,29	60	65,6	65,6	70,513	68,8
27	5055231***	95,61	77,14	82	71,43	65,6	91,026	78,7
28	5055231***	91,23	62,86	68	65,6	65,6	88,462	74,0
29	5055231***	88,6	62,86	78	65,6	67	76,923	72,4
30	5055231***	96,49	48,57	88	77,55	69	70,513	75,5
31	5055231***	92,98	77,14	88	83,67	76	84,615	83,2
32	5055231***	93,86	71,43	74	75,51	74	85,897	79,5
33	5055231***	90,35	80	86	67,35	70	88,462	78,0
34	5055231***	98,25	85,71	72	77,55	69	73,077	77,8
35	5055231***	92,11	60	62	67,35	65,6	84,615	73,1
36	5055231***	94,74	68,57	76	73,47	65,6	91,026	78,0
37	5055231***	94,74	77,14	82	67,35	66	91,026	77,7
38	5055231***	97,37	60	76	75,51	66	80,769	76,3
39	5055231***	93,42	45,71	80	65,6	65,6	76,923	71,6
40	5055231***	94,74	57,14	76	77,55	66	76,923	75,5

41	5055231***	98,68	74,29	70	65,6	65,6	83,333	75,1
42	5055231***	98,68	71,43	66	71,43	65,6	69,231	73,2
43	5055231***	98,68	68,57	78	81,63	65,6	80,769	78,8
44	5055231***	93,42	60	58	65,6	65,6	82,051	72,1
45	5055231***	96,05	62,86	62	65,6	65,6	75,641	71,7
46	5055231***	93,42	54,29	82	77,55	72	87,179	79,1
47	5055231***	94,74	60	68	65,6	65,6	76,923	72,0
48	5055231***	90,79	77,14	66	67,35	69	73,077	73,1

B. Reproduction Scores (2023 Cohort)

No.	NRP	PBL (20.00%)	Proyek Video (5.00%)	Kuis USG (10.00%)	Ujian Tengah Blok (25.00%)	UAB (30.00%)	Osce (10.00%)
1	5055231***	89,25	90	90	85	82	77,08
2	5055231***	89,75	90	90	79	76	72,92
3	5055231***	81,25	90	90	67	67	65,6
4	5055231***	83,5	90	70	66	66	65,6
5	5055231***	86	90	100	64	66	68,75
6	5055231***	86,25	90	100	71	66	72,92
7	5055231***	86,25	90	90	76	74	70,83
8	5055231***	87,25	90	90	69	70	66,67
9	5055231***	87,25	90	90	71	74	70,83
10	5055231***	89,75	90	100	77	82	68,75
11	5055231***	85,75	90	80	66	66	65,6
12	5055231***	84,75	90	90	67	73	68,75
13	5055231***	85	90	90	66	66	72,92
14	5055231***	84	90	90	66	81	75
15	5055231***	85	90	90	91	87	75
16	5055231***	84	90	90	66	72	
17	5055231***	87,25	90	100	87	87	66,67
18	5055231***	85	90	90	66	66	68,75
19	5055231***	85,5	90	100	71	72	68,75
20	5055231***	84,5	90	100	66	66	66,67
21	5055231***	87,75	90	90	74	78	75
22	5055231***	87,5	90	100	95	93	91,67
23	5055231***	83,25	90	90	67	68	70,83
24	5055231***	86,25	90	90	72	71	75
25	5055231***	83	90	100	68	75	68,75
26	5055231***	90,24	90	90	78	66	70,83
27	5055231***	85,64	90	100	79	77	77,08
28	5055231***	82,45	90	100	66	66	81,25
29	5055231***	74,45	90	100	73	75	66,67
30	5055231***	89,17	90	90	66	77	75

31	5055231***	84,05	90	100	82	81	70,83
32	5055231***	82,38	90	90	78	79	72,92
33	5055231***	81,19	90	100	79	79	75
34	5055231***	87,12	90	80	73	66	75
35	5055231***	83,19	90	90	66	66	77,08
36	5055231***	86,24	90	90	83	76	68,75
37	5055231***	85,05	90	80	66	77	70,83
38	5055231***	91,67	90	100	66	76	75
39	5055231***	86,67	90	90	75	66	79,17
40	5055231***	82,33	90	100	66	66	65,6
41	5055231***	93,33	90	100	80	75	79,17
42	5055231***	82,33	90	100	74	83	81,25
43	5055231***	86,67	90	100	84	70	72,92
44	5055231***	86,67	90	80	66	67	65,6
45	5055231***	82,67	90	90	66	66	68,75
46	5055231***	87,67	90	90	81	73	66,67
47	5055231***	83	90	90	66	66	65,6
48	5055231***	87	90	80	66	66	68,75

C. Respiration Scores (2023 Cohort)

No.	NRP	Kuis 1 (Written)	ETB (Written)	Kuis 2 (Written)	EAB (Written)	PBL	OSCE (Skill / Standard Patient)	Nilai Akhir
1	5055231***	84	82	72	92	89	79	85,27544
2	5055231***	92	74	64	84	90	79	82,3386
3	5055231***	78	66	48	66	88	73	73,59912
4	5055231***	66	66	66	80	89	75	77,37895
5	5055231***	70	66	52	66	87	69	72,30263
6	5055231***	82	80	74	76	89	88	83,34211
7	5055231***	78	74	66	78	89	88	81,67895
8	5055231***	88	74	58	72	89	77	78,49561
9	5055231***	82	72	68	82	89	73	79,72544
10	5055231***	88	84	76	92	90	85	87,5886
11	5055231***	84	72	48	70	89	81	78,09211
12	5055231***	80	82	44	76	89	71	78,54561
13	5055231***	84	80	62	80	90	85	83,4886
14	5055231***	68	66	58	70	91	81	77,11842
15	5055231***	94	88	74	84	93	88	88,19474
16	5055231***	78	74	52	66	82	77	74,65351
17	5055231***	88	88	76	86	94	94	89,90789
18	5055231***	68	66	46	66	93	73	74,57807
19	5055231***	94	74	50	80	91	77	80,78509

20	5055231***	70	66	66	74	89	98	81,22544
21	5055231***	82	76	74	82	93	94	86,04474
22	5055231***	88	90	78	92	93	96	91,7614
23	5055231***	82	76	56	82	92	83	82,79825
24	5055231***	74	82	60	66	91	81	79,91842
25	5055231***	82	72	68	72	92	75	78,93158
26	5055231***	74	66	50	72	88	77	75,51579
27	5055231***	74	70	64	76	88	90	80,33246
28	5055231***	82	78	60	74	88	90	81,73246
29	5055231***	80	72	60	80	89	73	78,56228
30	5055231***	70	66	72	66	90	83	77,27193
31	5055231***	82	90	78	86	89	90	87,95877
32	5055231***	82	74	62	72	88	90	80,63246
33	5055231***	88	74	66	86	89	90	84,45877
34	5055231***	90	70	60	74	90	92	81,7386
35	5055231***	78	70	66	76	88	92	81,04912
36	5055231***	88	76	72	90	88	90	85,43246
37	5055231***	82	74	62	72	89	85	80,06228
38	5055231***	82	74	74	82	87	77	80,4693
39	5055231***	80	72	52	70	82	79	75,30702
40	5055231***	84	62	58	68	82	73	72,15702
41	5055231***	80	68	52	82	84	98	81,44649
42	5055231***	78	78	42	70	84	90	78,77982
43	5055231***	62	84	58	86	84	94	84,01316
44	5055231***	72	68	54	72	82	81	75,02368
45	5055231***	68	66	54	74	82	79	74,40702
46	5055231***	84	70	78	82	82	92	81,30702
47	5055231***	82	70	48	76	79	81	75,63421
48	5055231***	62	66	64	74	84	75	74,56316

D. Emergency Scores (2023 Cohort)

No.	NRP	Presentasi telah kasus (10.00%)	Diskusi Kasus (10.00%)	Tugas Project Based / Case (20.00%)	Ujian Akhir Blok / Written test (30.00%)	Ujian OSCE (30.00%)
1	5055241***	85	82	79	82	72
2	5055241***	90	86	87	80	85
3	5055241***	90	90	94	78	79
4	5055241***	90	84	94	84	89
5	5055241***	90	84	87	86	76
6	5055241***	85	86	79	88	76

7	5055241***	85	86	79	72	81
8	5055241***	90	84	87	74	66
9	5055241***	90	80	94	88	84
10	5055241***	85	86	79	80	70
11	5055241***	90	84	87	80	94
12	5055241***	90	80	94	66	72
13	5055241***	90	86	89	76	68
14	5055241***	85	84	79	78	86
15	5055241***	85	84	79	68	82
16	5055241***	90	84	89	74	87
17	5055241***	90	84	87	84	89
18	5055241***	90	91	87	84	96
19	5055241***	90	95	89	84	79
20	5055241***	90	84	94	90	78
21	5055241***	90	95	94	82	92
22	5055241***	90	90	89	82	86
23	5055241***	90	80	89	80	87
24	5055241***	90	95	89	88	83
25	5055241***	85	95	79	86	92
26	5055241***	85	84	79	74	83
27	5055241***	90	84	87	82	90
28	5055241***	90	86	87	84	97
29	5055241***	90	86	94	80	82
30	5055241***	90	84	87	84	87
31	5055241***	85	84	79	82	73
32	5055241***	85	91	79	90	85
33	5055241***	90	95	87	94	86
34	5055241***	90	84	89	84	86
35	5055241***	85	84	79	68	88
36	5055241***	90	84	87	82	87
37	5055241***	90	84	94	86	80
38	5055241***	90	84	89	82	77
39	5055241***	90	84	87	92	78
40	5055241***	90	84	94	84	82
41	5055241***	90	84	89	80	80
42	5055241***	85	84	79	84	84
43	5055241***	90	84	94	76	92
44	5055241***	90	90	89	76	87
45	5055241***	90	84	94	84	92
46	5055241***	90	95	94	72	83
47	5055241***	90	90	89	84	80
48	5055241***	90	95	94	80	82
49	5055241***	85	95	79	90	82
50	5055241***	90	84	89	76	98

E. Clinic Technology A – Traklin A (2024 Cohort)

No	NRP	Nilai CBT (Written)	Nilai OSCE	Nilai Quiz (Case)	Nilai Tugas (Case)	Nilai Akhir
1	5055241***	72	85,95	82	95	82,53
2	5055241***	65,6	85,37	83	91	79,928
3	5055241***	65,6	86,04	88	94	81,396
4	5055241***	65,6	83,08	77	91	78,112
5	5055241***	78	89,66	89	94	86,714
6	5055241***	78	91,19	84	95	86,726
7	5055241***	64	84,69	93	97	81,576
8	5055241***	65,6	79,93	79	83	75,952
9	5055241***	70	84,6	82	93	81,09
10	5055241***	76	89,02	72	97	83,758
11	5055241***	80	86,47	87	91	85,288
12	5055241***	50	70,86	74	84	67,044
13	5055241***	65,6	73,49	83	90	75,026
14	5055241***	64	84,15	84	94	79,56
15	5055241***	72	90,46	94	95	86,134
16	5055241***	65,6	87,39	86	87	80,586
17	5055241***	72	85,03	71	97	80,812
18	5055241***	72	92,94	82	90	84,576
19	5055241***	76	81,53	84	72	78,812
20	5055241***	68	85,76	75	94	80,054
21	5055241***	74	83,7	80	96	82,08
22	5055241***	82	93,84	88	93	89,286
23	5055241***	68	88,32	85	94	82,578
24	5055241***	76	91,96	69	94	84,034
25	5055241***	80	89,93	93	96	88,322
26	5055241***	65,6	76,91	64	85	72,794
27	5055241***	74	89,97	94	96	86,688
28	5055241***	78	87,13	90	94	85,852
29	5055241***	65,6	79,6	77	93	77,02
30	5055241***	74	86,82	60	92	79,728
31	5055241***	65,6	88,57	84	95	81,958
32	5055241***	78	90,05	72	94	84,32
33	5055241***	78	92,83	93	96	88,882
34	5055241***	66	91,54	72	73	78,166
35	5055241***	62	88,37	80	94	80,048
36	5055241***	80	90,23	90	72	84,392
37	5055241***	70	84,97	98	90	83,188
38	5055241***	66	89,58	91	71	79,932
39	5055241***	74	84,54	84	91	82,266

40	5055241***	66	93,75	82	93	83,55
41	5055241***	65,6	83,21	93	94	81,014
42	5055241***	84	85,52	95	97	88,208
43	5055241***	68	88,37	84	90	81,848
44	5055241***	66	79,98	92	91	79,242
45	5055241***	65,6	85,56	91	92	81,354
46	5055241***	68	86,92	89	96	82,918
47	5055241***	72	79,91	92	94	81,464
48	5055241***	76	88,6	95	91	86,14
49	5055241***	76	91,85	91	96	87,59
50	5055241***	65,6	88,94	82	92	81,356

F. Clinic Technology B – Traklin A (2024 Cohort)

No	NRP	NILAI QUIZ (Cased Based)	NILAI TUGAS (Cased Based)	NILAI CBT (Written)	NILAI OSCE (Skill / SP)	Nilai Akhir
1	5055241***	80	97	72	77,37	79,948
2	5055241***	85	92	84	82,81	85,224
3	5055241***	90	98	84	83,97	87,388
4	5055241***	75	93	74	80,28	80,412
5	5055241***	90	95	76	79,97	82,788
6	5055241***	85	95	78	84,69	84,776
7	5055241***	95	98	48	82,94	76,676
8	5055241***	75	80	54	71,64	68,356
9	5055241***	85	95	78	85,91	85,264
10	5055241***	60	98	72	78,04	78,416
11	5055241***	85	93	74	87,87	84,448
12	5055241***	70	86	74	71,63	75,052
13	5055241***	80	91	62	69,06	72,424
14	5055241***	85	95	66	82,45	80,28
15	5055241***	95	97	70	83,75	83,4
16	5055241***	85	88	78	88,37	84,848
17	5055241***	65	98	68	88,57	81,928
18	5055241***	80	93	66	94,39	84,156
19	5055241***	85	62	70	78,72	73,388
20	5055241***	75	97	84	77,57	83,128
21	5055241***	80	100	82	88,84	88,136
22	5055241***	85	97	84	87,76	88,204
23	5055241***	90	95	70	86,48	83,592
24	5055241***	60	98	82	90,23	86,292
25	5055241***	95	97	80	90,47	89,088

26	5055241***	55	83	62	86,6	75,34
27	5055241***	95	100	80	90,15	89,56
28	5055241***	95	98	74	91,17	87,768
29	5055241***	70	93	68	83,93	79,572
30	5055241***	60	92	90	81,89	84,156
31	5055241***	85	95	54	79,25	75,4
32	5055241***	70	93	76	79,55	80,22
33	5055241***	95	95	78	90,26	88,004
34	5055241***	50	65	76	86,52	75,408
35	5055241***	75	98	66	83,93	80,472
36	5055241***	90	63	76	84,3	78,12
37	5055241***	100	92	78	84,54	85,616
38	5055241***	90	65	70	78,53	74,412
39	5055241***	80	93	60	79,9	76,56
40	5055241***	85	95	80	88,49	86,896
41	5055241***	95	98	86	82,88	88,052
42	5055241***	100	98	76	86,1	86,84
43	5055241***	80	92	74	91,39	85,156
44	5055241***	90	92	74	84,66	83,464
45	5055241***	95	95	74	88,64	86,156
46	5055241***	95	97	74	86	85,5
47	5055241***	95	95	74	83,17	83,968
48	5055241***	95	95	78	86,95	86,68
49	5055241***	95	98	74	87,1	86,14
50	5055241***	75	97	82	93,73	88,992

APPENDIX C: DETAILED RESULTS

This appendix presents supplementary statistical results that support Chapter 4. All calculations are based on the Python script described in Appendix A and the pre-processed data described in Section 4.1.

A. Clustering quality metrics (k=3)

Metric	Value	Interpretation
Silhouette Coefficient	0.3913	Moderate to acceptable separation
Calinski-Harabasz Index	61.70	Well-defined clusters
Davies-Bouldin Index	0.9195	Good cluster compactness (lower is better)
Cophenetic Correlation	0.6401	Moderate hierarchical fit

B. Full ANOVA Table

Modality	F-statistic	p-value	Significance
Written Theoretical	30.51	<0.001	***
Practical Skill	95.11	<0.001	***
Standardised Patient	95.11	<0.001	***
Problem Based	54.67	<0.001	***

C. 3 Raw Score Means and Standard Deviations per Cluster

Cluster	Statistic	Written Theoretical	Practical Skill	Standardised Patient	Problem Based
1 (At-Risk)	Mean Std	68.2 10.1	62.3 11.8	62.3 11.8	84.1 9.2
2 (Mixed)	Mean Std	81.5 8.4	79.8 9.1	79.8 9.1	52.7 14.3
3 (Average)	Mean Std	76.8 7.9	83.2 8.5	83.2 8.5	88.4 7.1

D. Standardised Cluster Profiles

Cluster	Written Theoretical	Practical Skill	Standardised Patient	Problem Based	Overall Mean
1 (At-Risk)	-1.035	-1.358	-1.358	-0.055	-0.952
2 (Mixed)	+0.574	+0.399	+0.399	-2.236	-0.216
3 (Average)	+0.345	+0.503	+0.503	+0.342	+0.424

E. Cluster Sizes and Percentages

Cluster	Label	Frequency	Percentage
1	(At-Risk)	26	26.5%
2	(Mixed)	9	9.1%
3	(Average)	63	64.3%
Total		98	100%

AUTHOR'S BIOGRAPHY



The author was born in Tangerang, Indonesia, on November 20, 2004. He completed his primary education at Future International School, Al Ain, United Arab Emirates. He then continued his elementary and junior high school education at ADNOC Schools, Abu Dhabi, United Arab Emirates, before completing his senior high school education at SMA Lazuardi Depok, Indonesia.

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